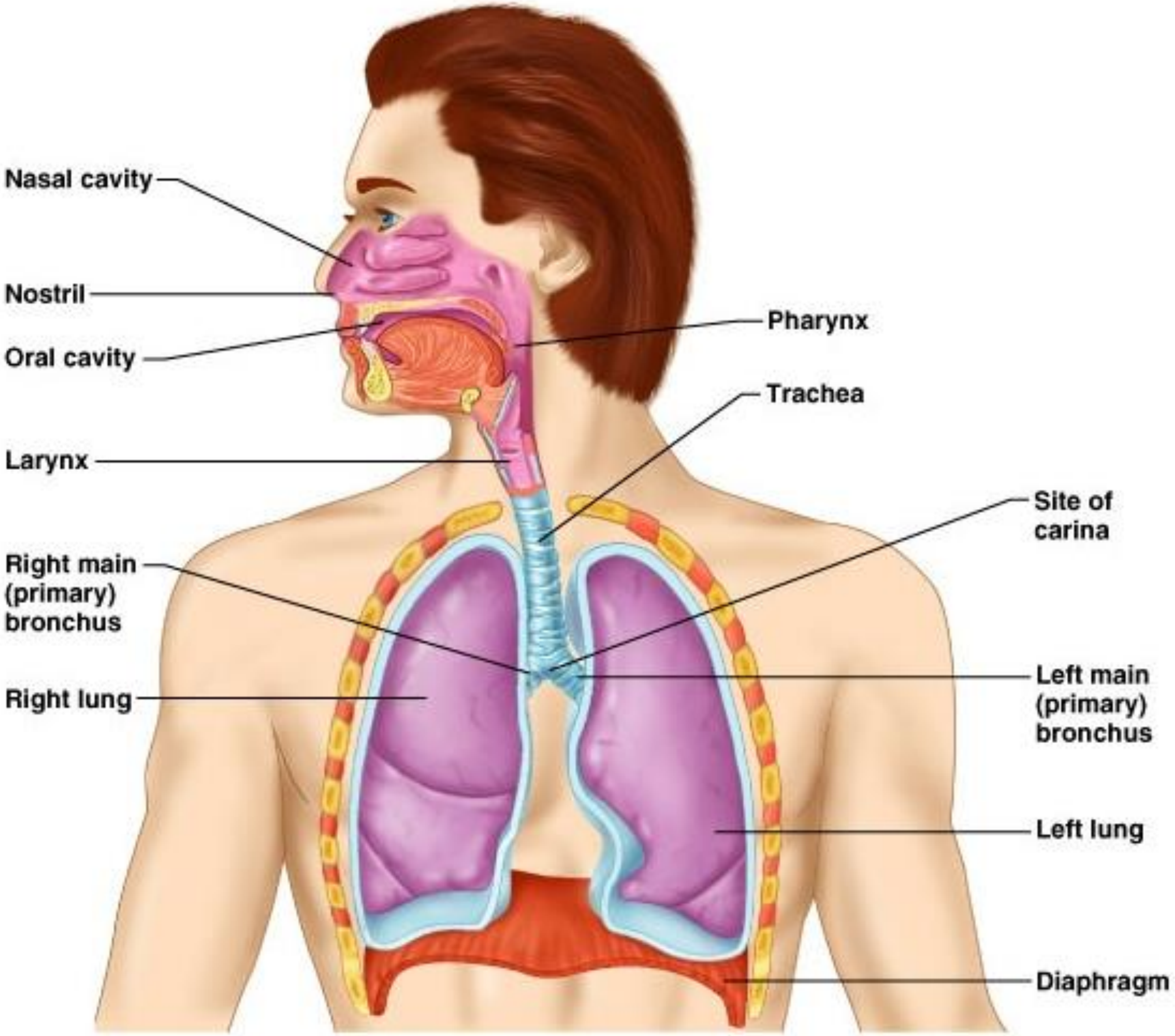
Respiratory system

By: Fikre Bayu (MSc)

The Respiratory System



Respiration processes

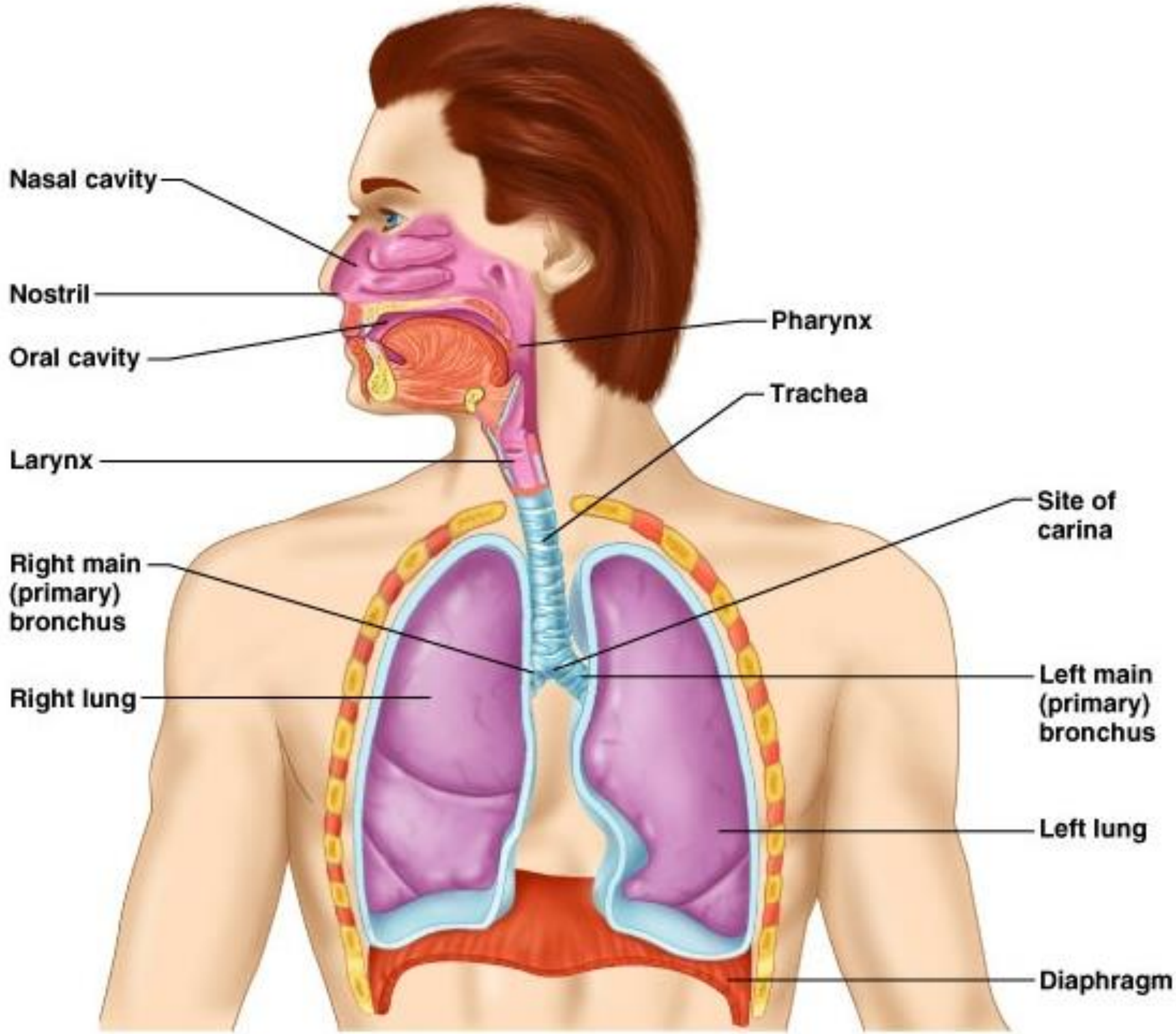
- **Pulmonary ventilation**
  - Air moves in and out of lungs
  - Continuous replacement of gases in alveoli (air sacs)
- **External respiration**
  - Gas exchange between blood and air at **alveoli**
  - $\mathbf{O_2}$ in air diffuses into blood and $\mathbf{CO_2}$ in blood diffuses into air
- **Transport of respiratory gases**
  - Between the **lungs** and the **cells** of the body
  - Performed by the **cardiovascular system**
- **Internal respiration**
  - Gas exchange in **capillaries** b/n blood and tissue cells
  - $\mathbf{O_2}$ in blood diffuses into tissues and $\mathbf{CO_2}$ waste in tissues diffuses into blood

Respiratory System Functions

- **Gas conditioning**:
  - Warmed
  - Humidified
  - Cleaned of particles
- **Sound production**:
  - Movement of air over **true vocal cords**
  - Also involves **nose, Paranasal sinuses, teeth, lips and tongue**
- **Olfaction**:
  - Olfactory epithelium over **superior nasal conchae**
- **Defense**:
  - Course hairs (Vibrissae), mucus, lymphoid tissue

Respiratory System

- The respiratory system includes the **lungs** and a **system of tubes** that links the sites of gas exchange with the external environment



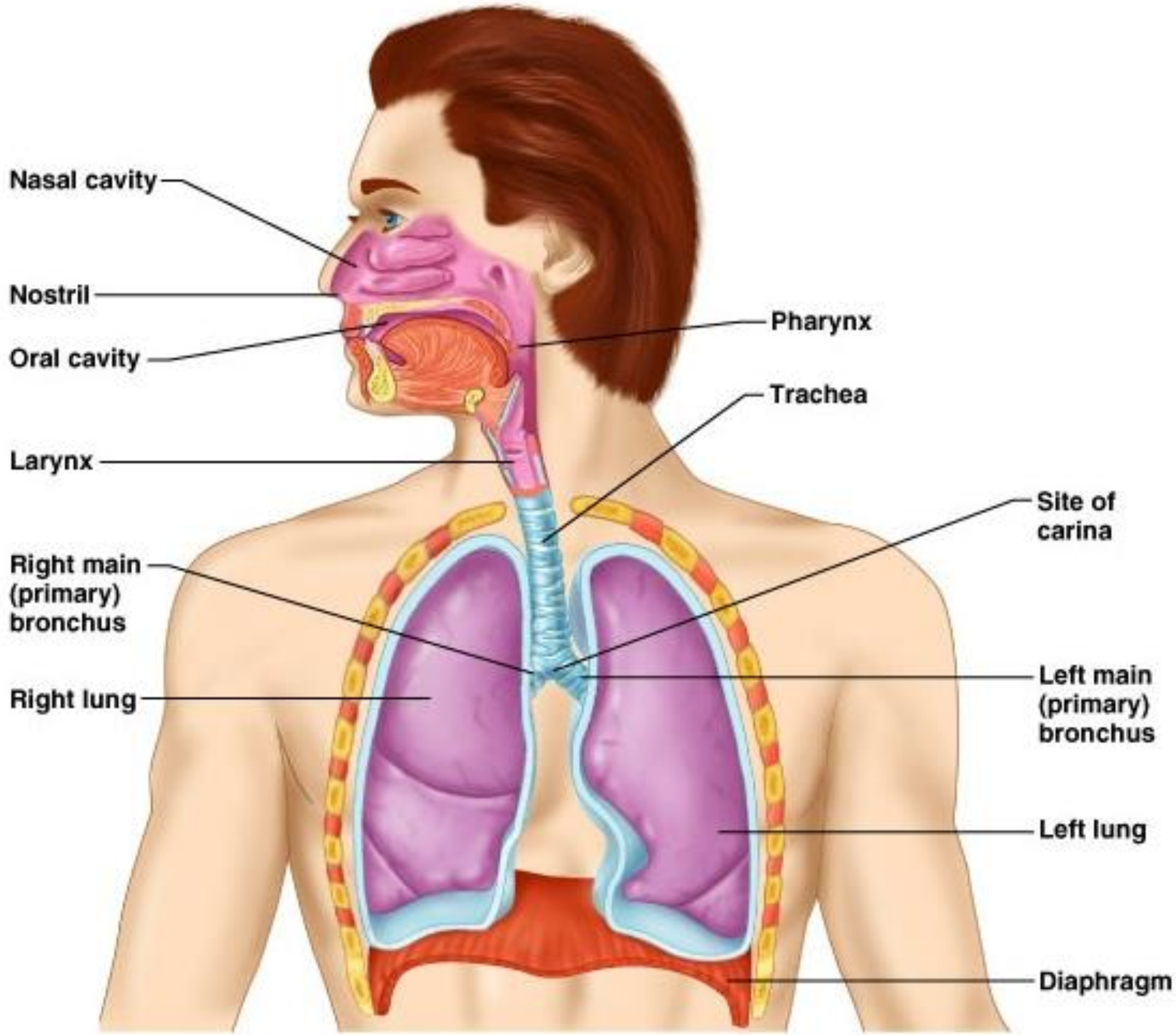
The Functional classification of RS

1. Conducting zone

- Respiratory passages that carry air to the site of gas exchange
- **Filters, humidifies** and **warms air**

2. Respiratory zone

- **Site of gas exchange**
- Composed of
  - Respiratory bronchioles
  - Alveolar ducts
  - Alveolar sacs



Cont….

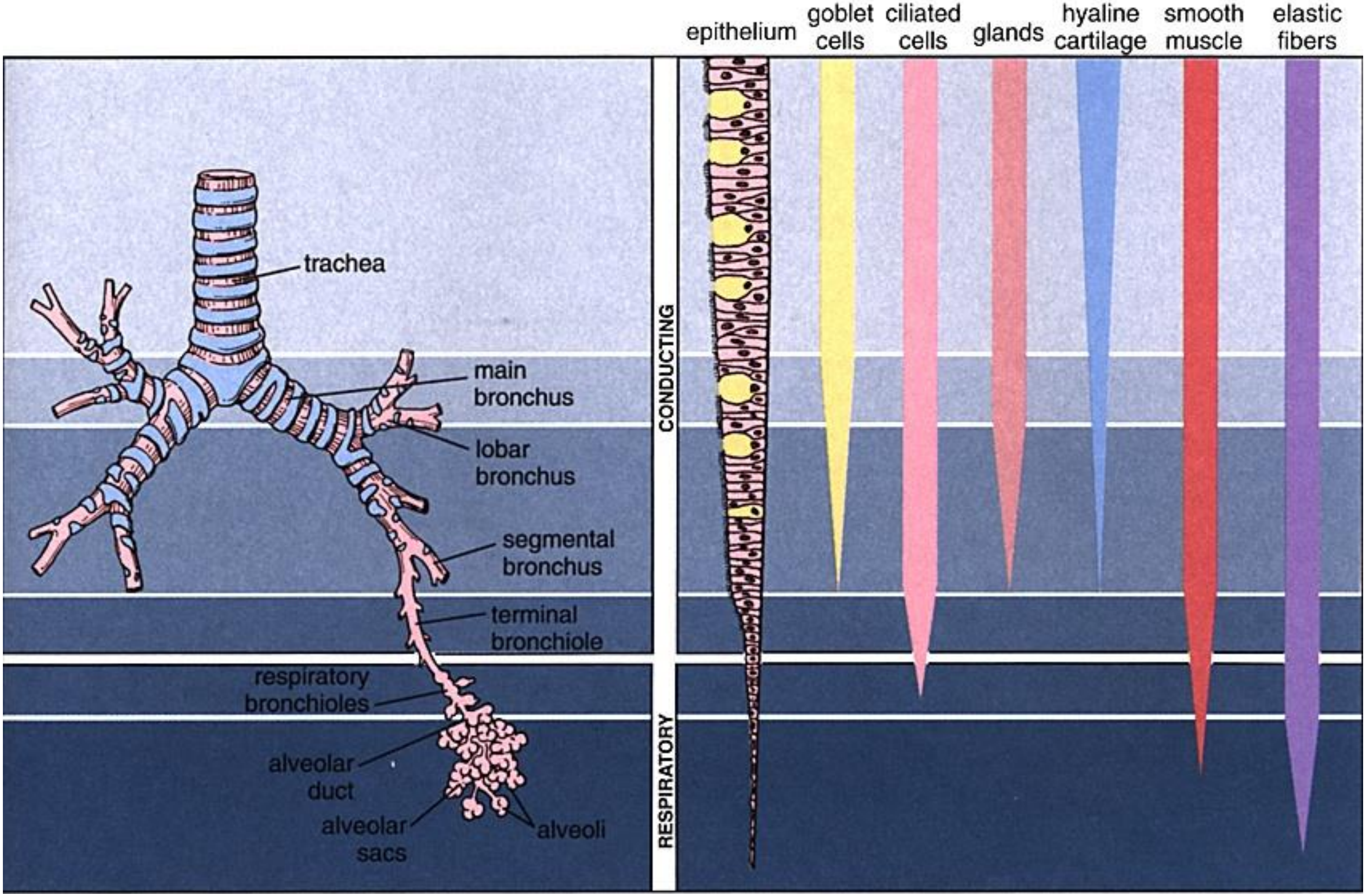
- The **conducting portion** serves two main functions:

1. to provide a ***conduit*** through which air can travel to and from the lungs and to **condition the inspired air**

2. to ensure an ***uninterrupted supply of air***, a combination of

- **cartilage ➔rigid structural support**
- **elastic and collagen fibers➔ flexibility**
- **smooth muscle➔ extensibility**
- .

Cont….

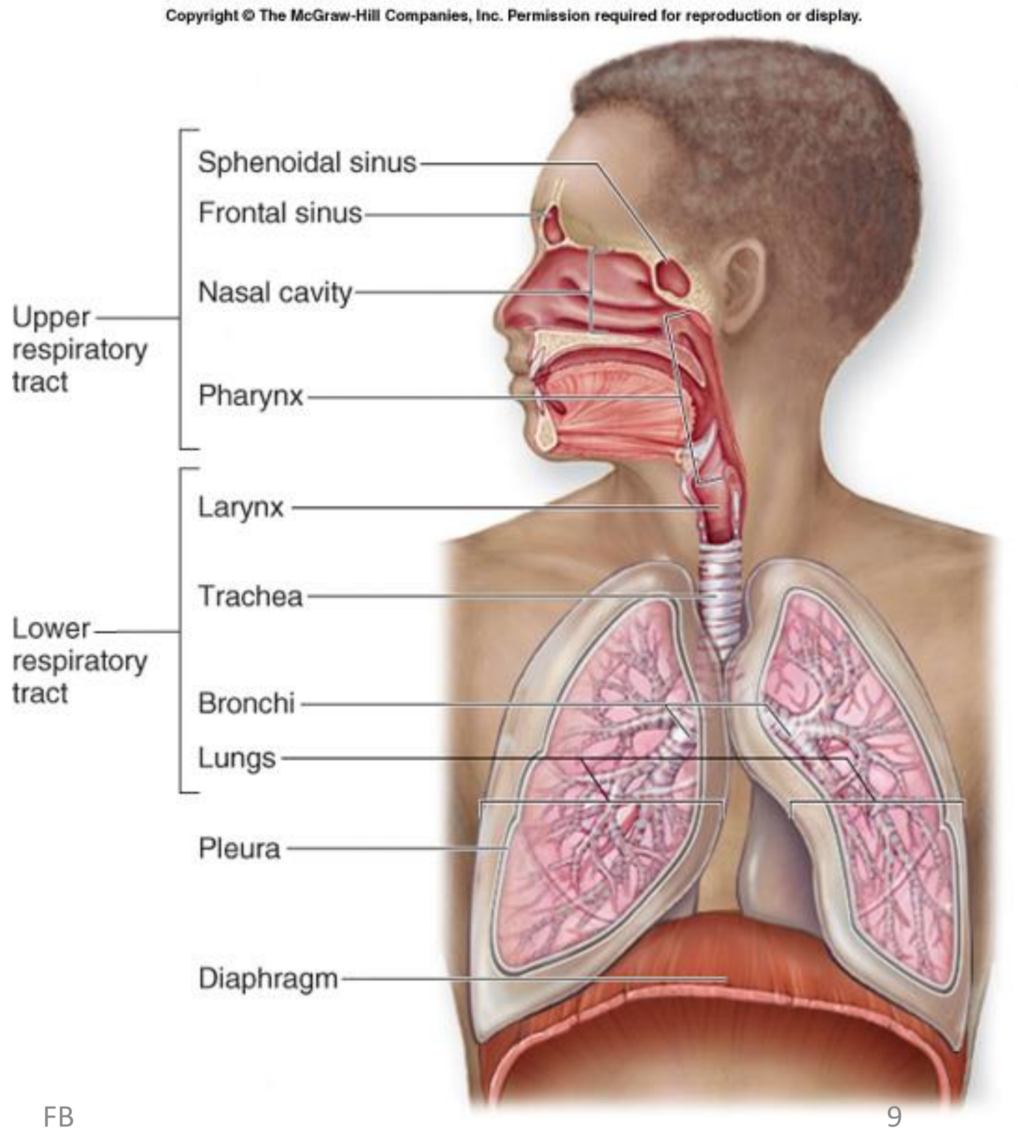


epithelium
goblet cells
ciliated cells
glands
hyaline cartilage
smooth muscle
elastic fibers
trachea
main bronchus
lobar bronchus
segmental bronchus
terminal bronchiole
respiratory bronchioles
alveolar duct
alveolar sacs
alveoli
CONDUCTING
RESPIRATORY

Structural classification of RS

1. Upper respiratory tract

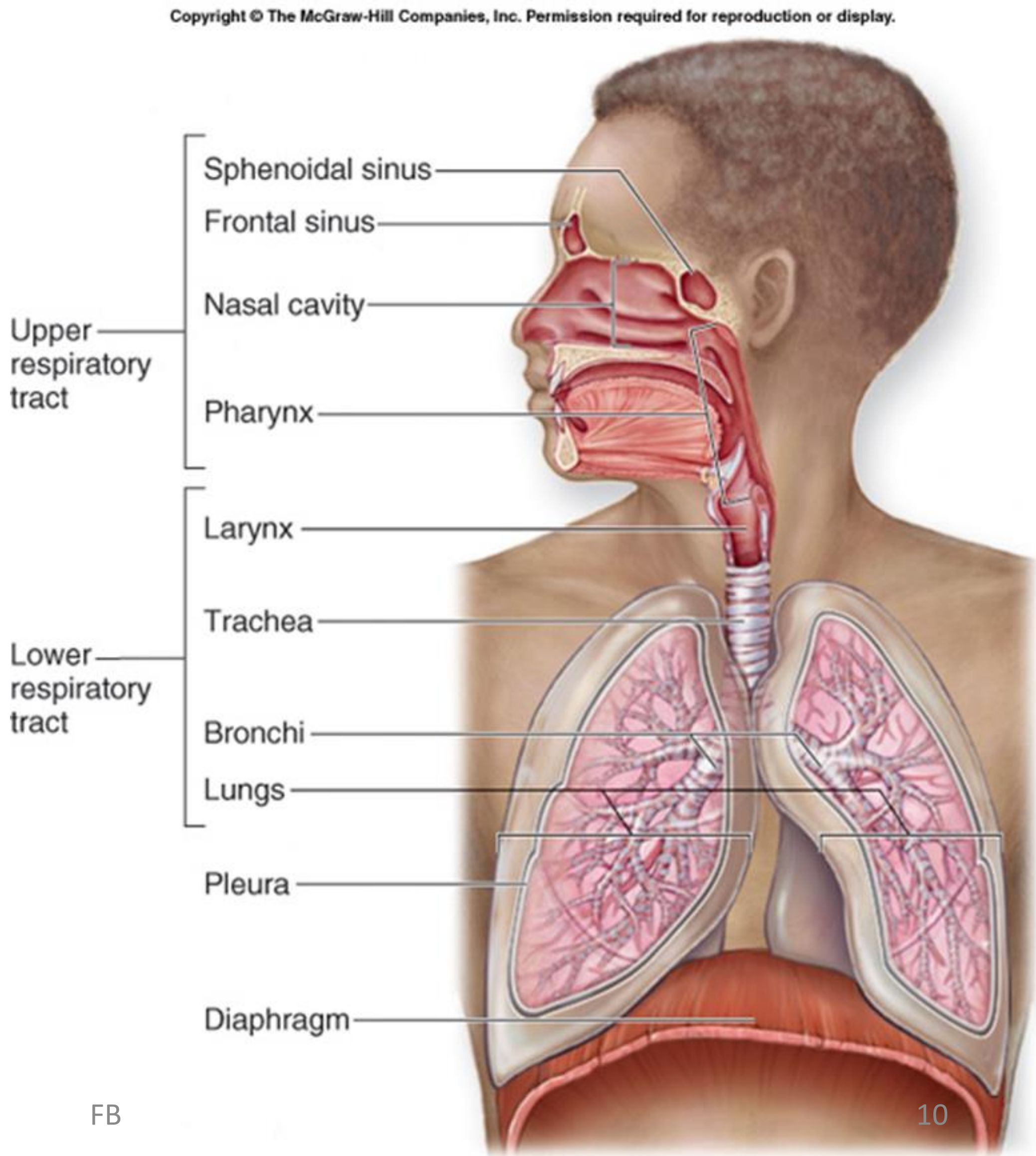
- Composed of
  - **the nose**
  - **the nasal cavity**
  - **the Paranasal sinuses**
  - **the pharynx** (throat)
  - and associated structures
- All part of the **conducting** portion of the respiratory system



Cont.…

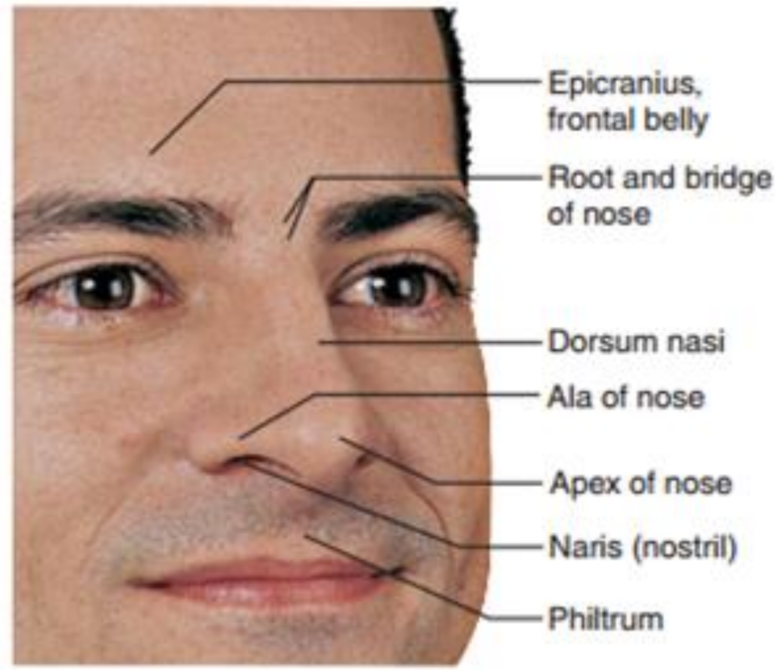
2. Lower respiratory tract

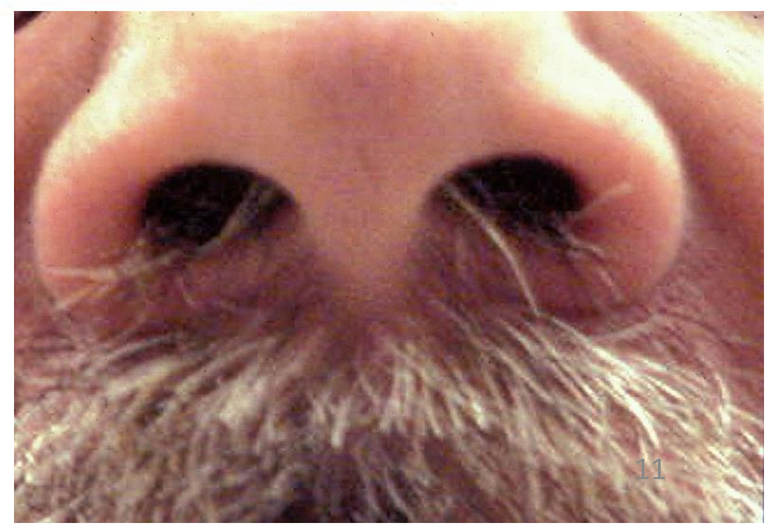
- Includes:
  - **Larynx**
  - **Trachea**
  - **Bronchi**
  - **Bronchioles**
  - **Lungs**



Upper Respiratory Tract: Nose

- It is the only **externally visible** part of the respiratory system
  - ➢ provides an airway for respiration
  - ➢ moistens and **warms** entering air
  - ➢ **filters** inhaled air to cleanse it of foreign particles
  - ➢ serves as a **resonating chamber** for speech
  - ➢ houses the **olfactory (smell) receptors**
- It is divided into **two** regions
  - – **The external nose**, including the root, bridge, dorsum nasi, and apex
  - – **The internal nasal cavity**





Nasal Cavity

- It **lies in** and **posterior** to the external nose
- Air passes through the **external nares or nostrils**
- The nasal cavity is divided into right and left halves by the **nasal septum** in the midline; formed by the **perpendicular plate** of the ethmoid bone, the **Vomer,** and a **septal cartilage**
- Posteriorly, the nasal cavity is continuous with the nasal part of the pharynx through the posterior nasal apertures, also called the **choanae** or **internal nares**
- The **roof** is formed by **ethmoid** and **sphenoid** bones
- The **floor** is formed by the **hard and soft** palates

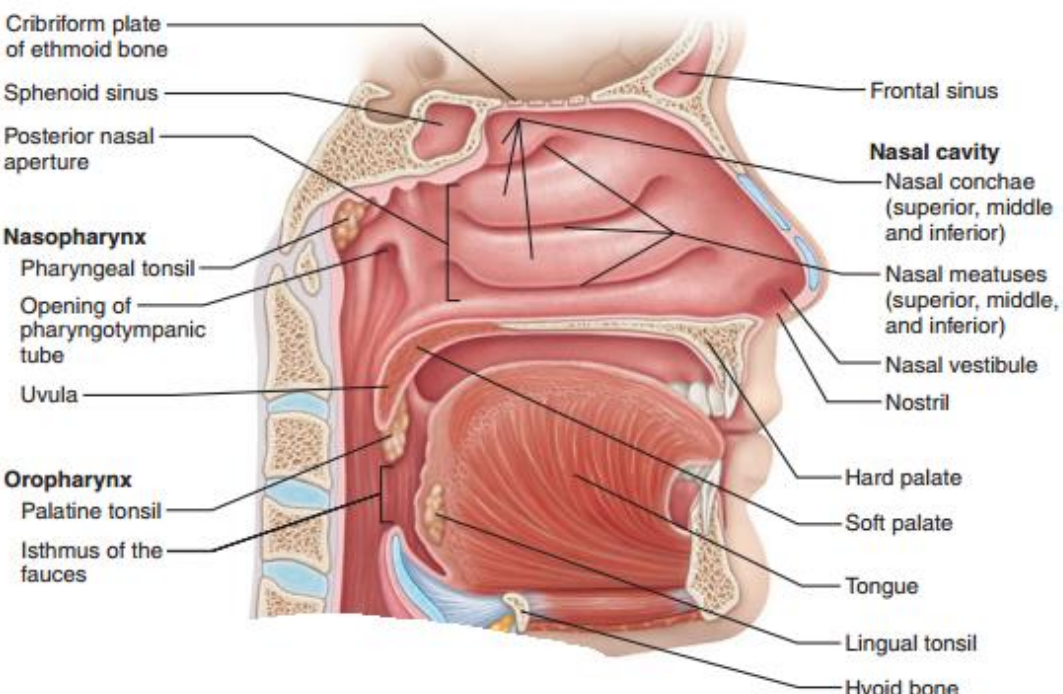
Cont.…

❖**Superior, medial, and inferior conchae**:

– Protrude medially from the lateral walls

– Increase mucosal area

– Enhance air turbulence and help filter air



Cont….

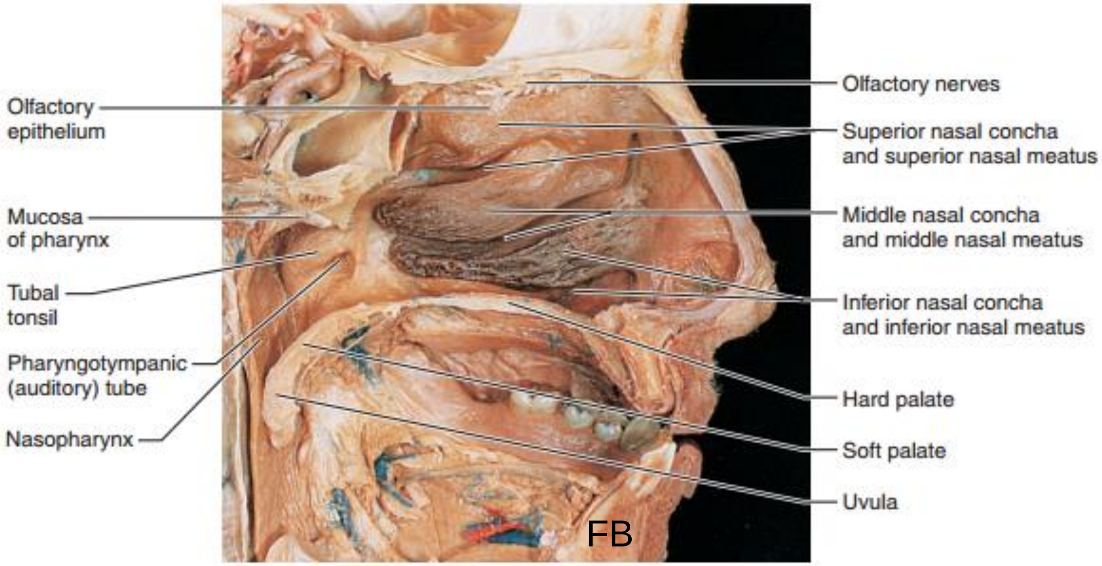
- **Vestibule** – nasal cavity superior to the nares lined with skin containing sebaceous and sweat glands
  - This contains **hairs (Vibrissae)** that filter coarse particles from inspired air

1. Olfactory mucosa

- Lines the **superior nasal cavity**
- Contains **smell receptors**

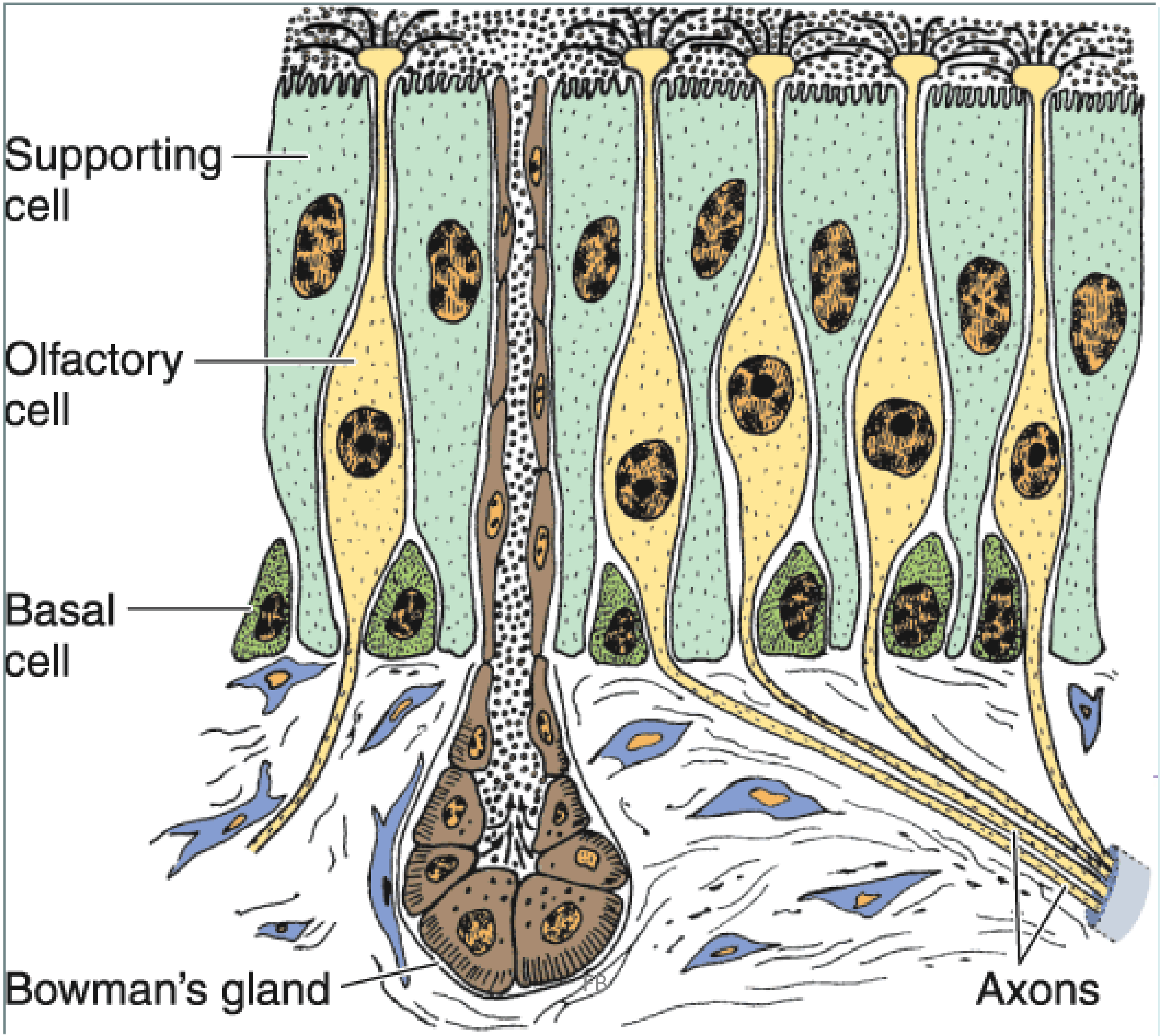
2. Respiratory mucosa

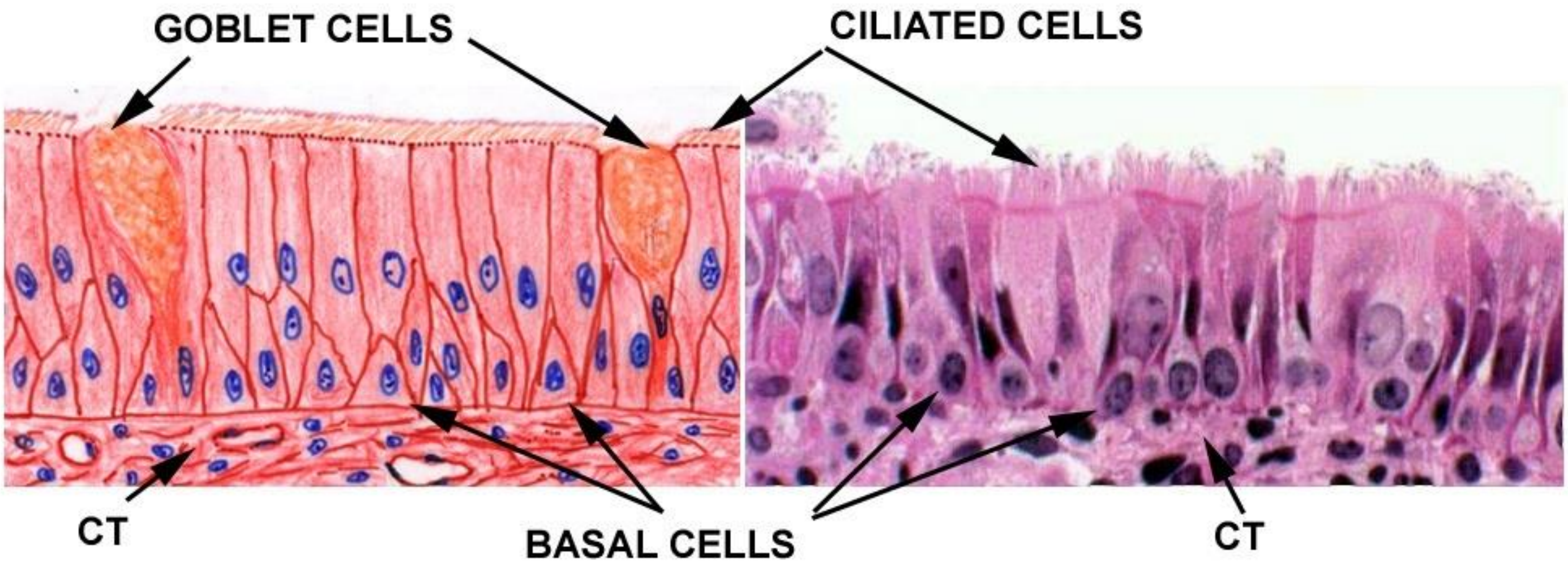
- **lines the vast majority of the nasal cavity**

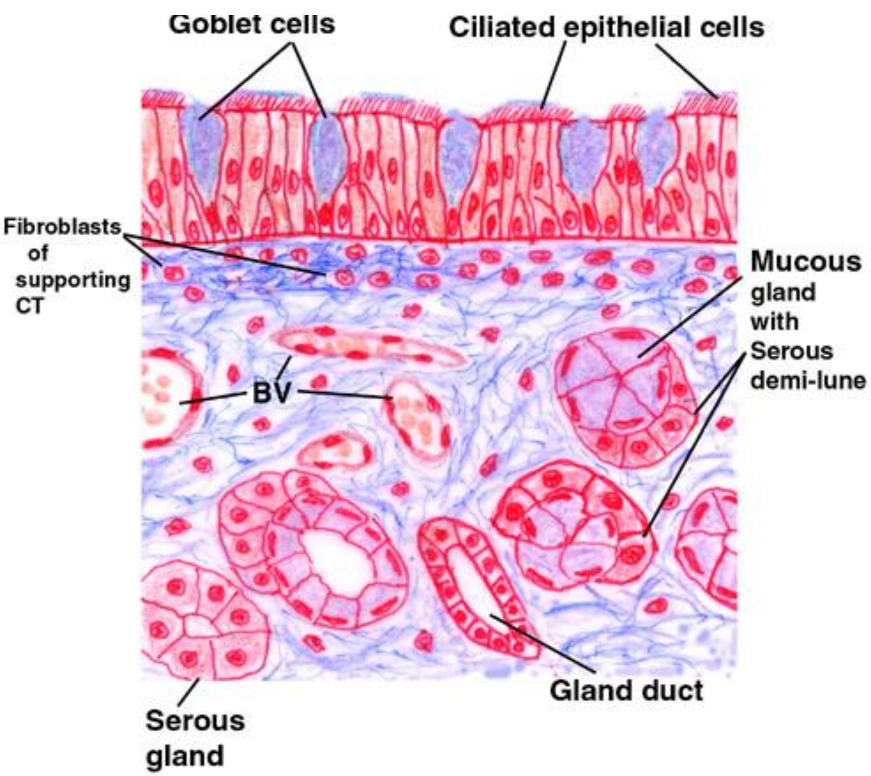


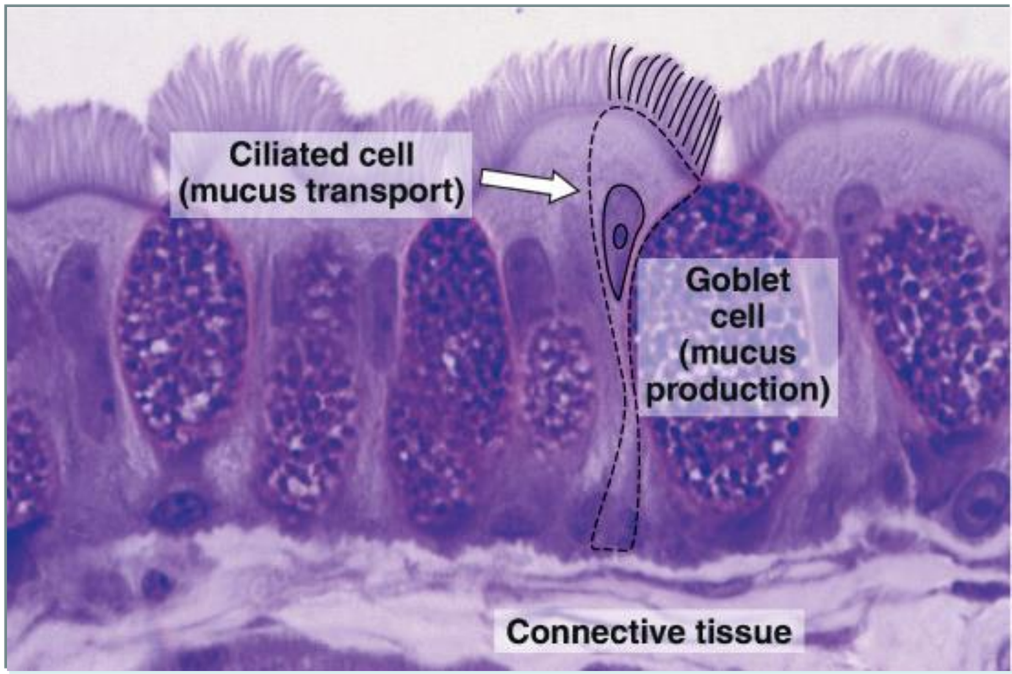
FB

Olfactory mucosa





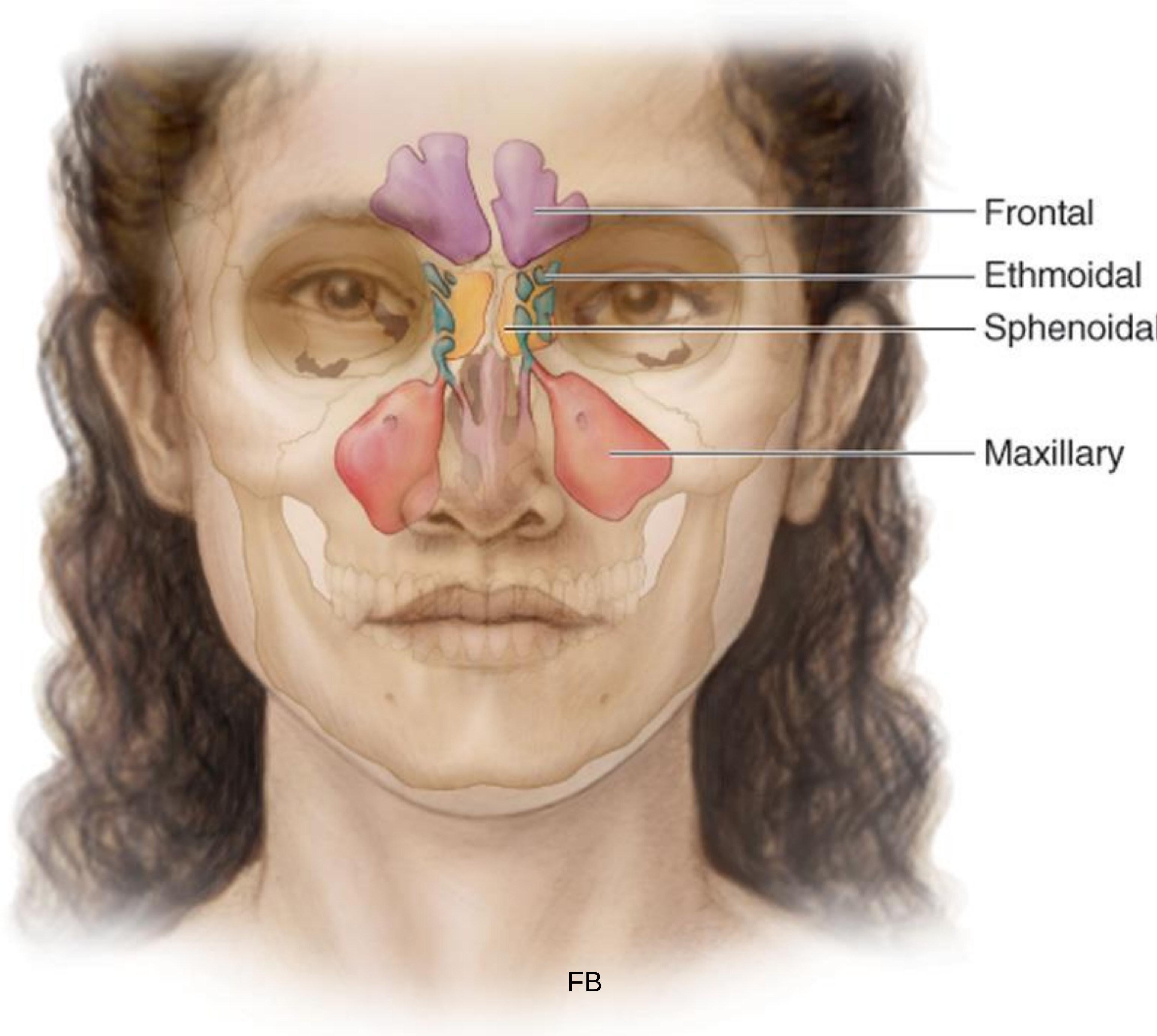




Photomicrograph illustrating the main components of the respiratory epithelium. Pararosaniline–toluidine blue (PT) stain. High magnification.

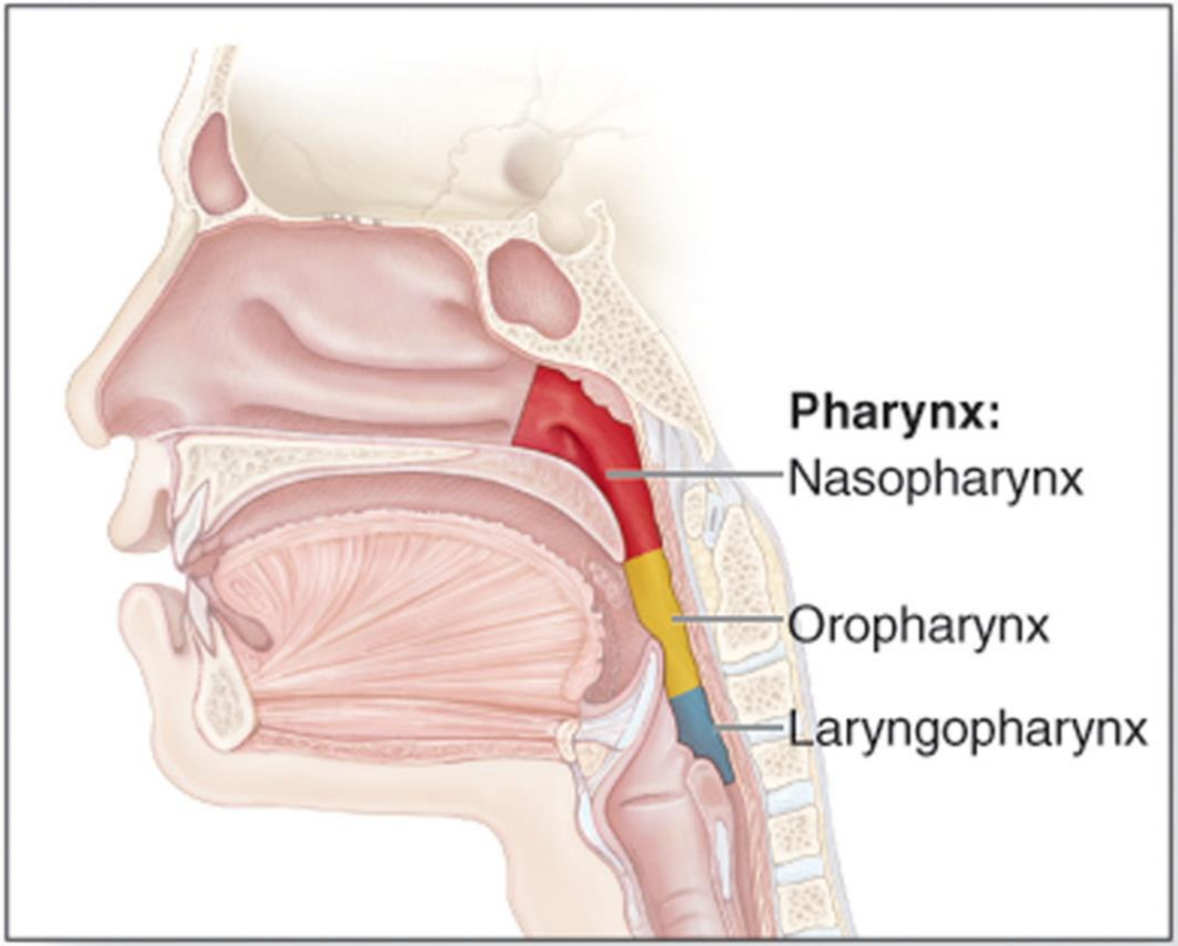
Paranasal Sinuses

- **Paranasal sinuses:**
  - In four skull bones
  - paired air spaces
  - **decrease skull bone weight**
  - **Produce resonating sound**
- Named for the bones in which they are housed
  - frontal
  - ethmoidal
  - sphenoidal
  - maxillary
- Communicate with the nasal cavity by ducts
- Covered with the same Pseudostratified ciliated columnar epithelium as the nasal cavity


Frontal
Ethmoidal
Sphenoidal
Maxillary

Pharynx

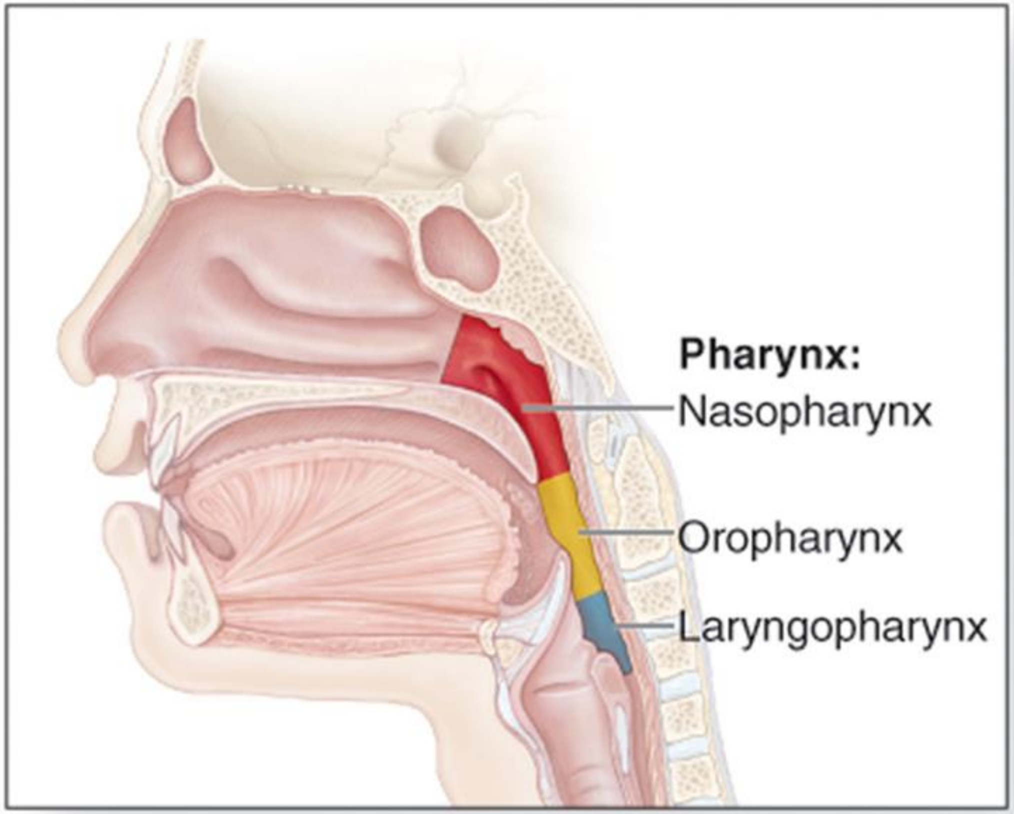
- Commonly called the **throat**
- Funnel-shaped
  - **slightly wider superiorly and narrower inferiorly**
- Originates posterior to the nasal and oral cavities
- Extends inferiorly near the level of the bifurcation of the **larynx** and **esophagus**
- Common pathway for both **air** and **food**



(b) Regions of pharynx

Cont….

- **Flexible walls**
  - lined by a mucosa
  - contain skeletal muscles primarily used for swallowing
  - distensible
  - to force swallowed food into the esophagus
- **Partitioned into three adjoining regions:**
  - nasopharynx
  - oropharynx
  - laryngopharynx



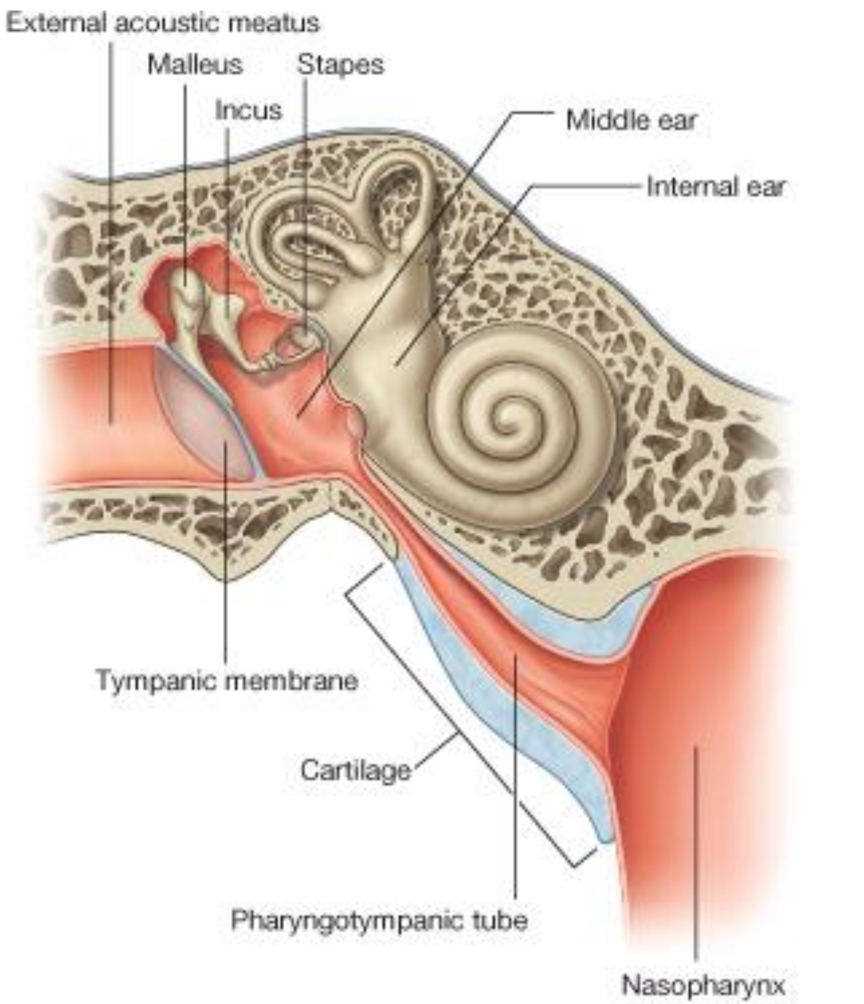
(b) Regions of pharynx

Nasopharynx

- Superior most region of the pharynx
- Lined by **pseudostratified ciliated columnar epithelium**
- **Location:**
  - posterior to the nasal cavity
  - superior to the soft palate
    - separates it from the posterior part of the oral cavity
- Normally, only **air** passes through
- **Soft palate(Uvula)**
  - Blocks material from the oral cavity and oropharynx
  - elevates when we swallow

Cont….

- **Auditory tubes**
  - paired
  - In the lateral walls of the nasopharynx
  - connect the nasopharynx to the middle ear
- **Pharyngeal tonsil**
  - posterior nasopharynx wall
  - single
  - commonly called the **adenoids**



Drake et al: Gray's Anatomy for Students - www.studentcon

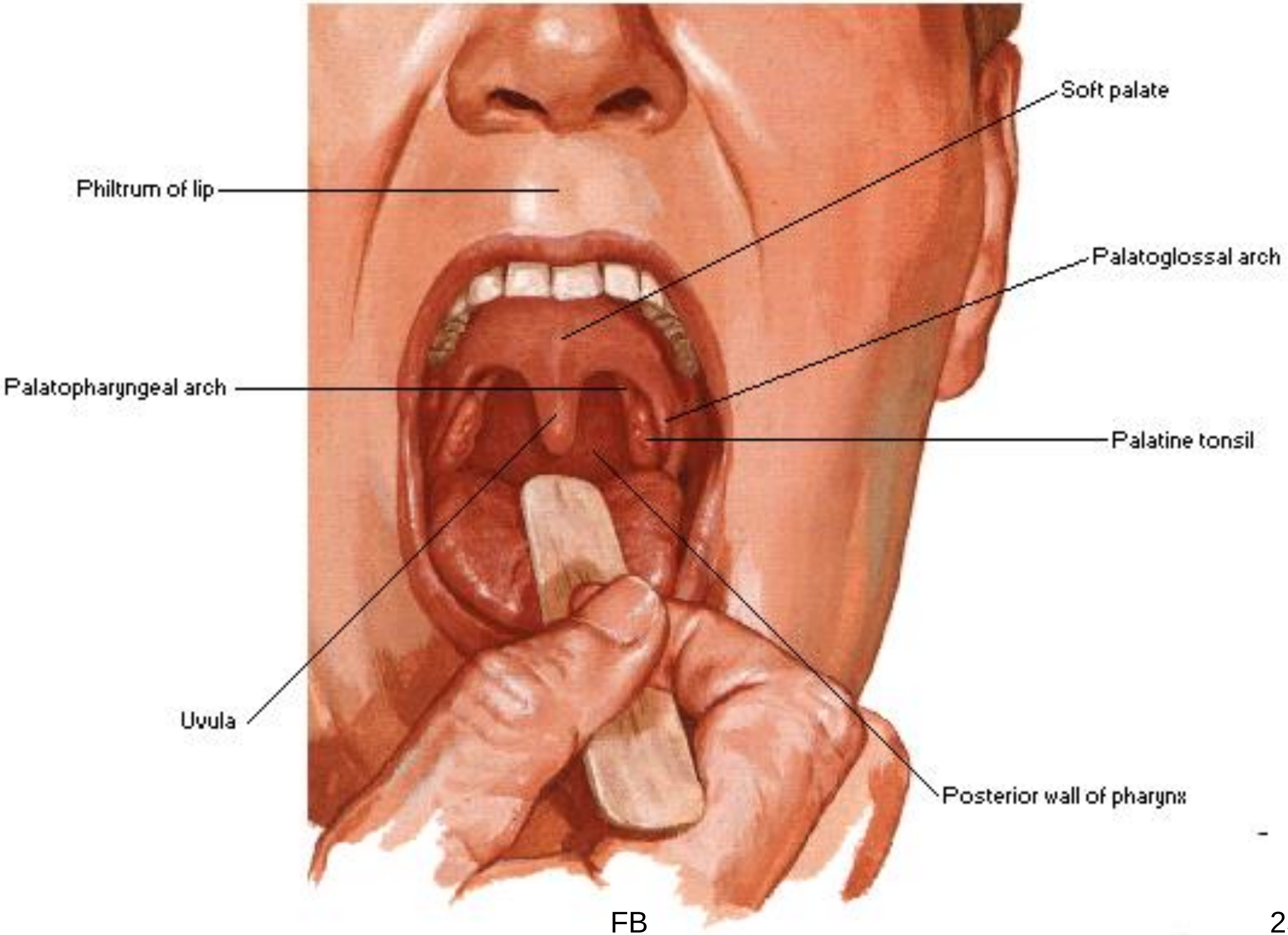
Oropharynx

- The middle pharyngeal region
- **Location:**
  - Immediately posterior to the oral cavity
  - Lined by **non keratinized stratified squamous epithelium**
  - Bounded by the soft palate superiorly,
  - the hyoid bone inferiorly
- Common respiratory and digestive pathway
  - both air and swallowed food and drink pass through

Cont….

- 2 pairs of muscular arches
  - anterior palatoglossal arches
  - posterior palatopharyngeal arches
  - form the entrance from the oral cavity.
- Lymphatic organs
  - provide the "first line of defense" against ingested or inhaled foreign materials.
  - Palatine tonsils
    - on the lateral wall between the arches
  - Lingual tonsils
    - At the base of the tongue

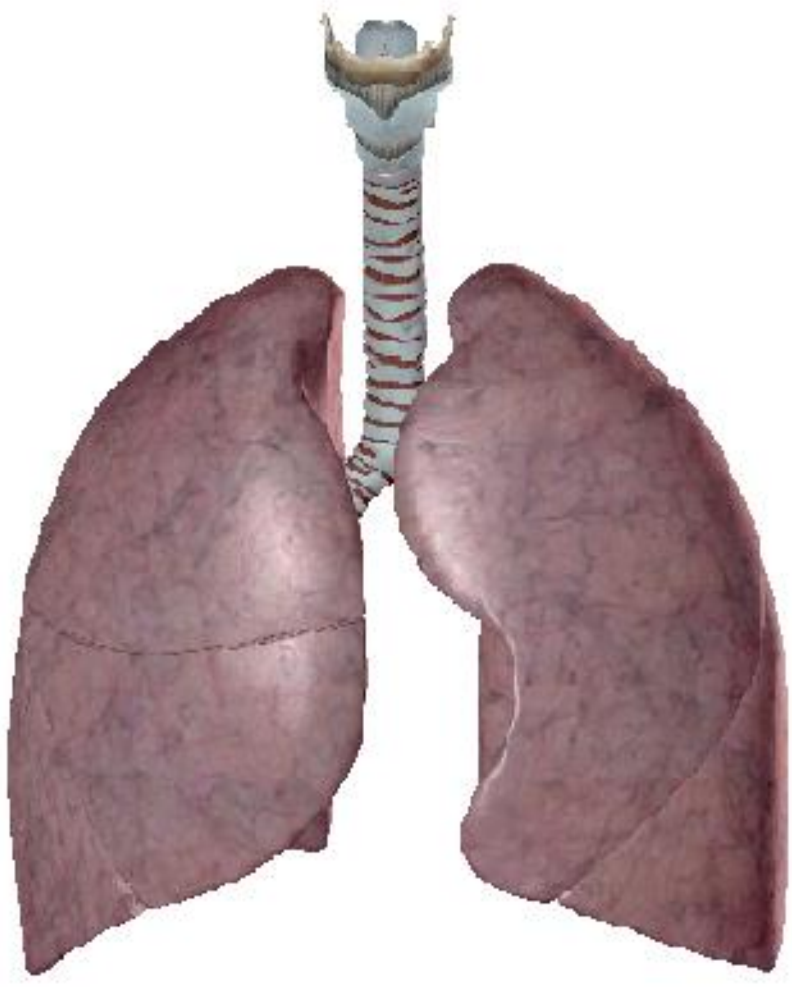
Dorsum of Tongue and Palate



Laryngopharynx

- Inferior, narrowed region of the pharynx
- **Location:**
  - Extends inferiorly from the **hyoid** bone
  - is continuous with the **larynx** and **esophagus**
  - Terminates at the superior border of the esophagus
    - is equivalent to the inferior border of the cricoid cartilage in the larynx

Lower Respiratory Tract



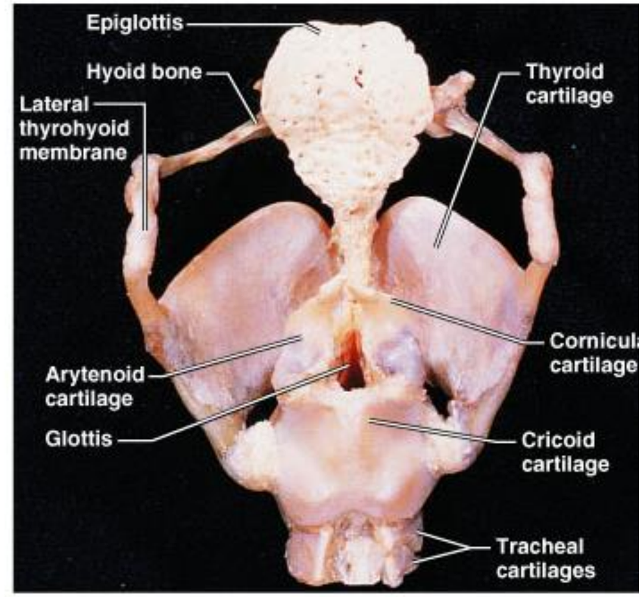
Larynx (voice box)

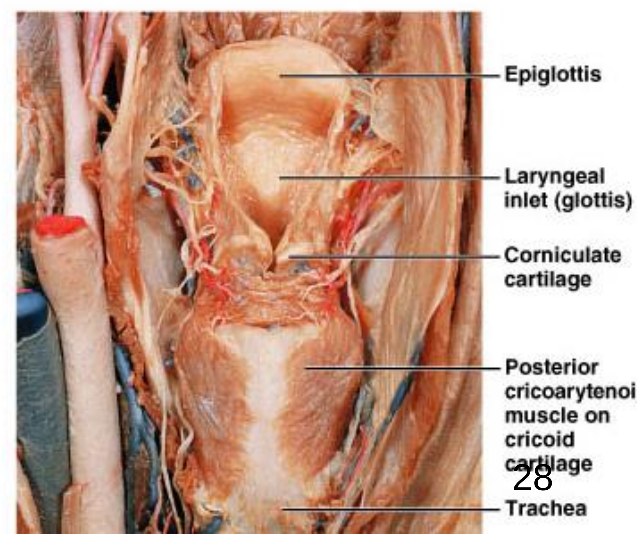
- Extends from the level of the 4th to the 6th cervical vertebrae
- Attaches to hyoid bone superiorly and Inferiorly is continuous with trachea (windpipe)

Functions:

1. Produces vocalizations (speech)
2. Provides an open airway (breathing)
3. Switching mechanism to route air & food into proper channels
   - Closed during swallowing
   - Open during breathing

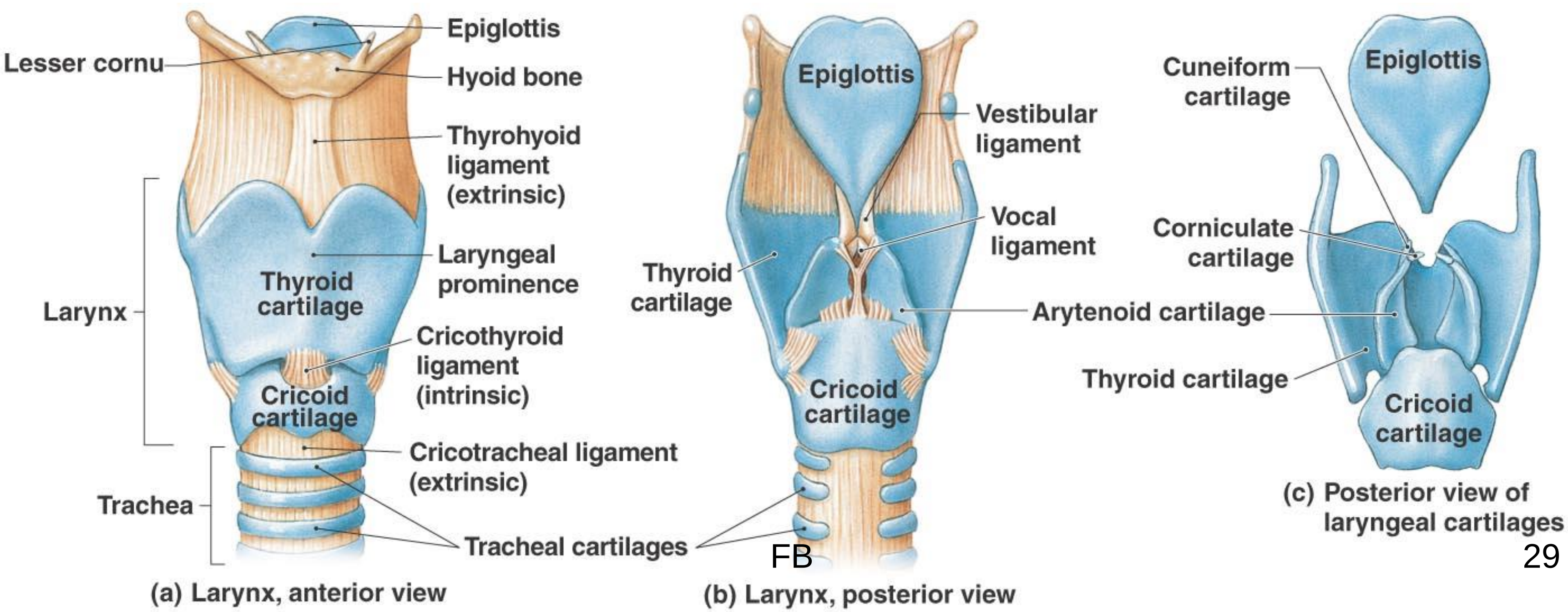
(c)





Framework of the larynx

- **Nine** pieces of cartilage
  - three individual pieces(**Thyroid**, **Cricoid**, **Epiglottis** cartilages)
  - three cartilage pairs (**Arytenoids**: on cricoid, **Corniculates**: attach to arytenoids, **Cuneiforms**: in aryepiglottic fold
  - held in place by **ligaments and muscles**
    - Intrinsic muscles: **regulate tension on true vocal cords**
    - Extrinsic muscles: **stabilize the larynx**



Cont.…

❑ Thyroid cartilage

- The **largest** cartilage in the larynx
- Made fused plates of **two hyaline** cartilages
- Forms **anterior wall of larynx**
- The anterior junction of the two plates forms the **laryngeal prominence (Adam's apple)**

❑ Epiglottis

- Large leaf shaped piece of **elastic** cartilage covered with epithelium
- Has a "stem" and "leaf"
- The stem is attached to the **anterior rim of thyroid cartilage**
- The broad superior leaf is unattached and free to move up and down
- The epiglottis prevents food and water from entering the airways

Cont.…

❖ Cricoid cartilage

- **The only complete ring of hyaline** cartilage
- It attaches to the first ring of trachea by **cricotracheal** ligament

❑ Arytenoid cartilages

- Its body is **hyaline** cartilage while the tips are **elastic cartilage**
- Located above **cricoid cartilage**
- They attach to the **true vocal cords** and **pharyngeal muscles** and function in **voice production**

❑ Corniculate cartilages

- Small horn shaped **elastic** cartilages
- Found at the **apex of each arytenoid cartilage**

❑ Cuneiform cartilages

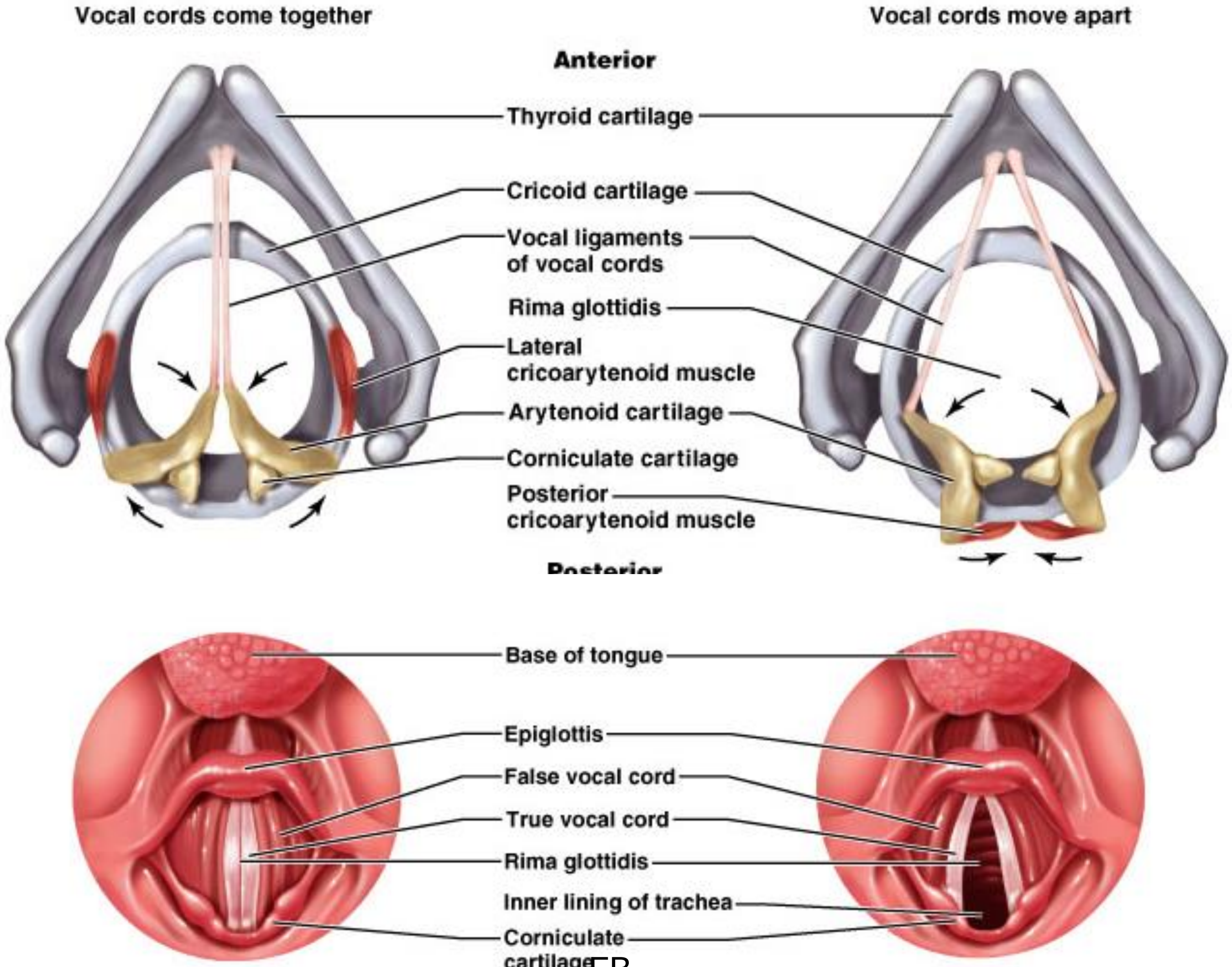
- Found in **aryepiglottic fold** and composed of **elastic cartilage**
- Support the **vocal folds and lateral aspects of the epiglottis**

Mechanism of voice production

- True vocal cords contain elastic ligaments stretched between pieces of rigid cartilage
- Intrinsic laryngeal muscles attach to both the cartilages and the true vocal cords **when the muscles contract**, **they pull the elastic ligaments tight** which moves the **true vocal cords out into the air** passage way
- The air pushing against the true vocal cords causes them to vibrate and thus creates sound
- Sound is created by the vibration of the vocal cords but the other structures are useful in changing these sounds into recognizable speech
- Pitch is controlled by the tension of the true vocal cords

  -if stretched tightly ,they vibrate more rapidly resulting in a high pitch and lower pitch sounds are produced by decreased muscular tension

Rima glottidis: opening between the vocal folds
Glottis: rima glottidis and the vocal folds

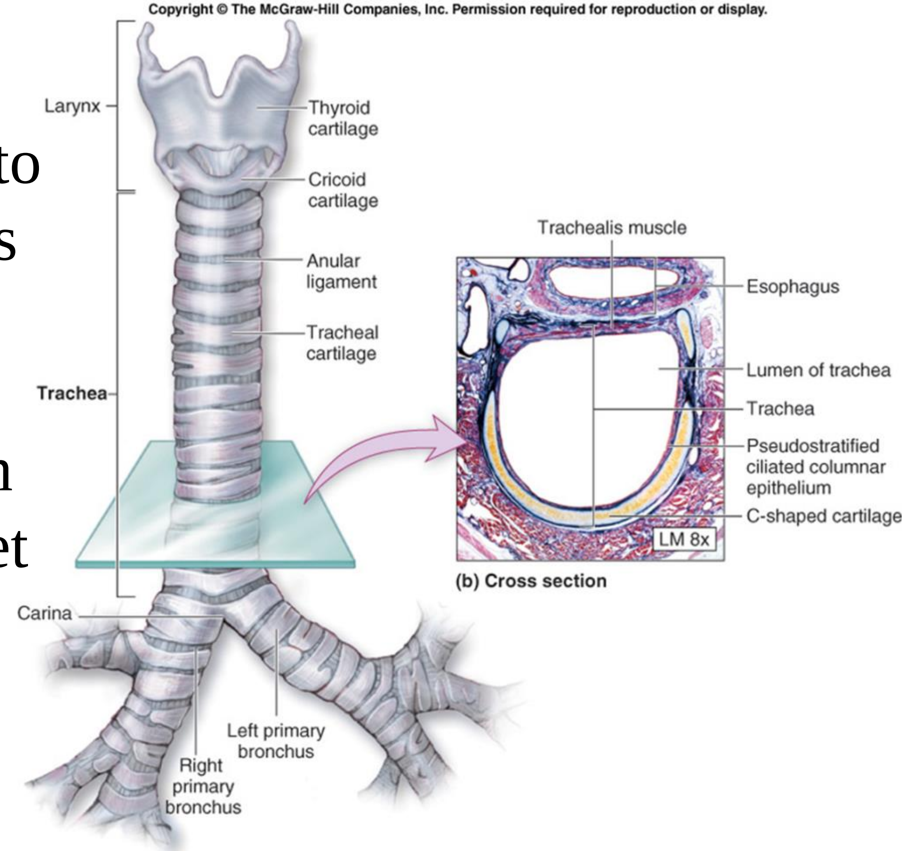


Trachea (Wind pipe)

- Tubular passageway for air located **anterior to the esophagus**
- Extends from **larynx** to the upper part of the **fifth thoracic vertebra(T5)**
- Then it divides into **right** and **left primary bronchi**
- Lined by mucous membrane composed of **pseudostratified ciliated columnar epithelium (PCCE)**
- The cilia here move mucus and trapped particles up to the pharynx
- The cartilage layer consists **15(16)-20 C-shaped rings of hyaline** cartilage stacked one on top of another by **annular** ligament

Cont….

- **Trachealis smooth muscle** complete the diameter of trachea posteriorly
  - ✓Esophagus can expand when food swallowed
  - ✓Food can be forcibly expelled
- Wall of trachea has layers common to many tubular organs – filters, warms & moistens incoming air
  - **Mucous membrane**:- pseudostratified epithelium with cilia & lamina propria with sheet of elastin
  - **Submucosa**:- with seromucous glands
  - **Adventitia:-** connective tissue which contains tracheal cartilages

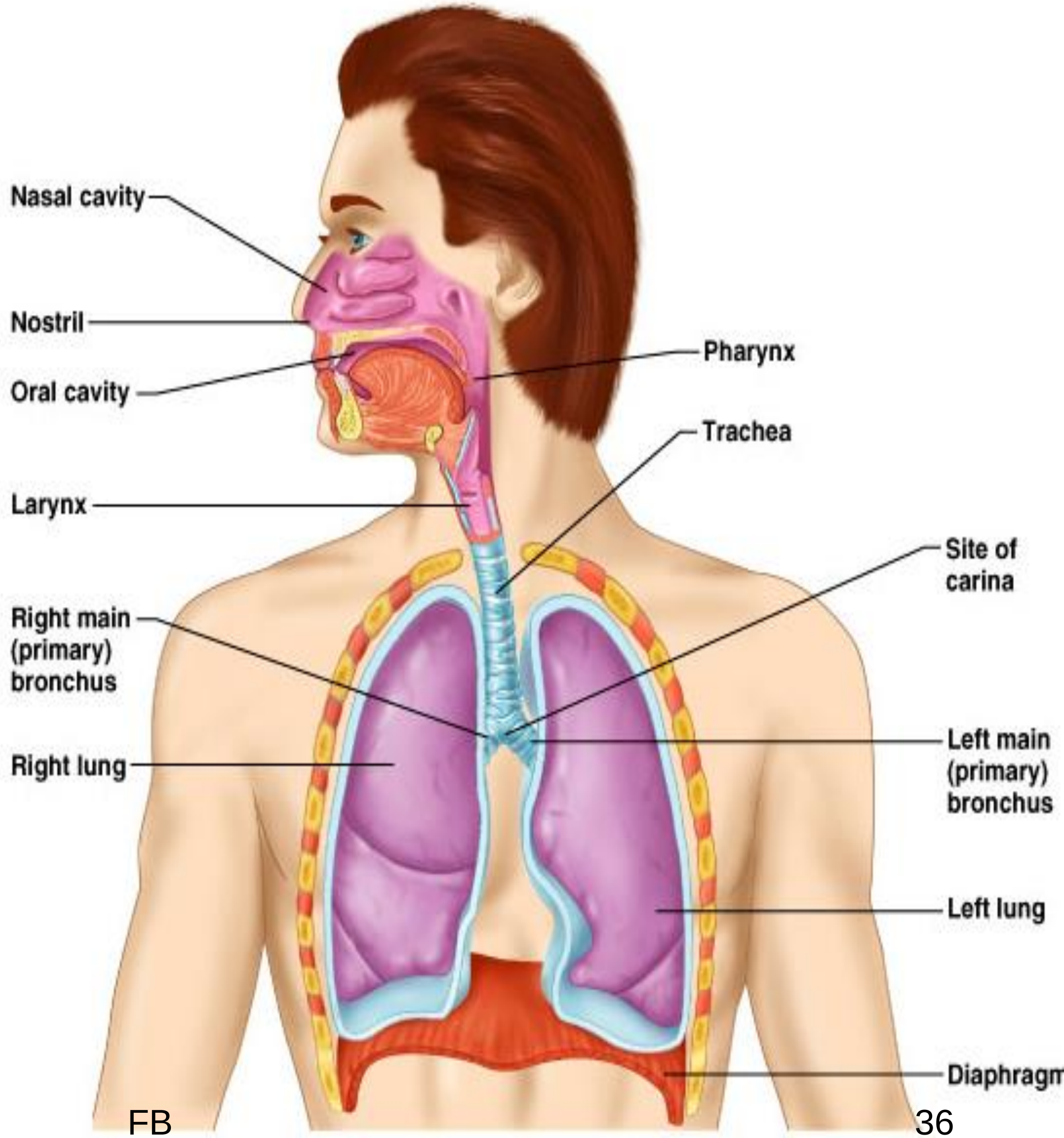


(a) Anterior view

(b) Cross section

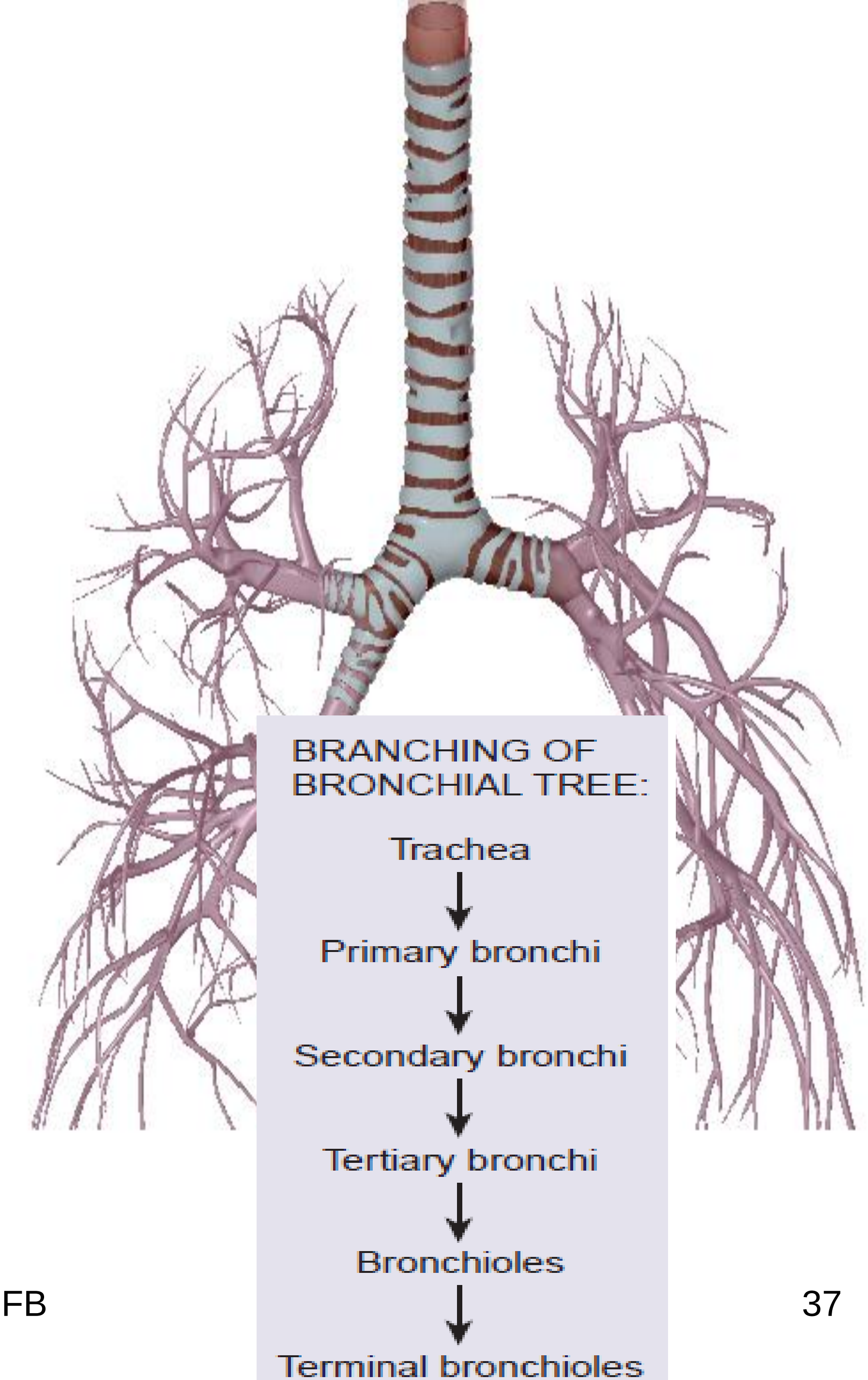
Carina

- Ridge on internal aspect of last tracheal cartilage
- Point where trachea branches (when alive & standing is at T7)
- Mucosa highly sensitive to irritants: cough reflex



Bronchial Tree

- A highly branched system
  - air-conducting passages
  - originate from the left and right primary bronchi.
- Progressively branch into narrower tubes as they diverge throughout the lungs before terminating in terminal bronchioles



Cont….

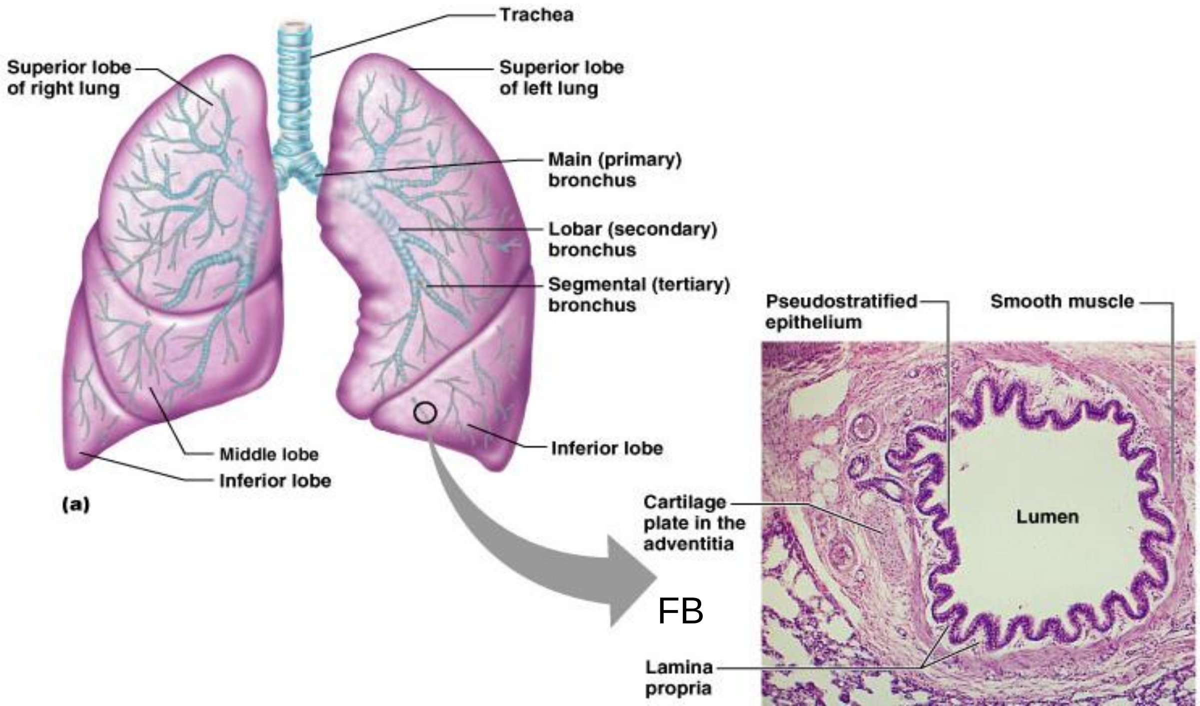
- Primary bronchi
  - Incomplete rings of hyaline cartilage ensure that they remain open
  - Right primary bronchus
    - shorter, wider, and more vertically oriented than the left primary bronchus
  - Foreign particles are more likely to lodge in the right primary bronchus
- Primary bronchi
  - enter the hilum of each lung
  - Also entering hilum:
    - pulmonary vessels
    - lymphatic vessels and nerves

Cont….

- **Secondary bronchi (or lobar bronchi)**
  - Branch of primary bronchus
  - left lung:
    - **two lobes**
    - two secondary bronchi
  - right lung
    - **three lobes**
    - three secondary bronchi
- **Tertiary bronchi (or segmental bronchi)**
  - Branch of secondary bronchi
  - left lung is supplied by **8 to 10** tertiary bronchi.
  - right lung is supplied by **10** tertiary bronchi
  - supply a part of the lung called a **Bronchopulmonary segment**
  - Surgically removed

Cont….

- Tubes smaller than 1 mm called bronchioles
- Smallest, terminal bronchioles, are less than 0.5 mm diameter
- Tissue changes as becomes smaller
  - **Cartilage plates, not rings, then disappears**
  - **Pseudostratified columnar to simple columnar to simple cuboidal without mucus or cilia**
  - **Smooth muscle important:**
    a) Sympathetic relaxation:-“**bronchodilation**”
    b) Parasympathetic constriction **:-“bronchoconstriction**”



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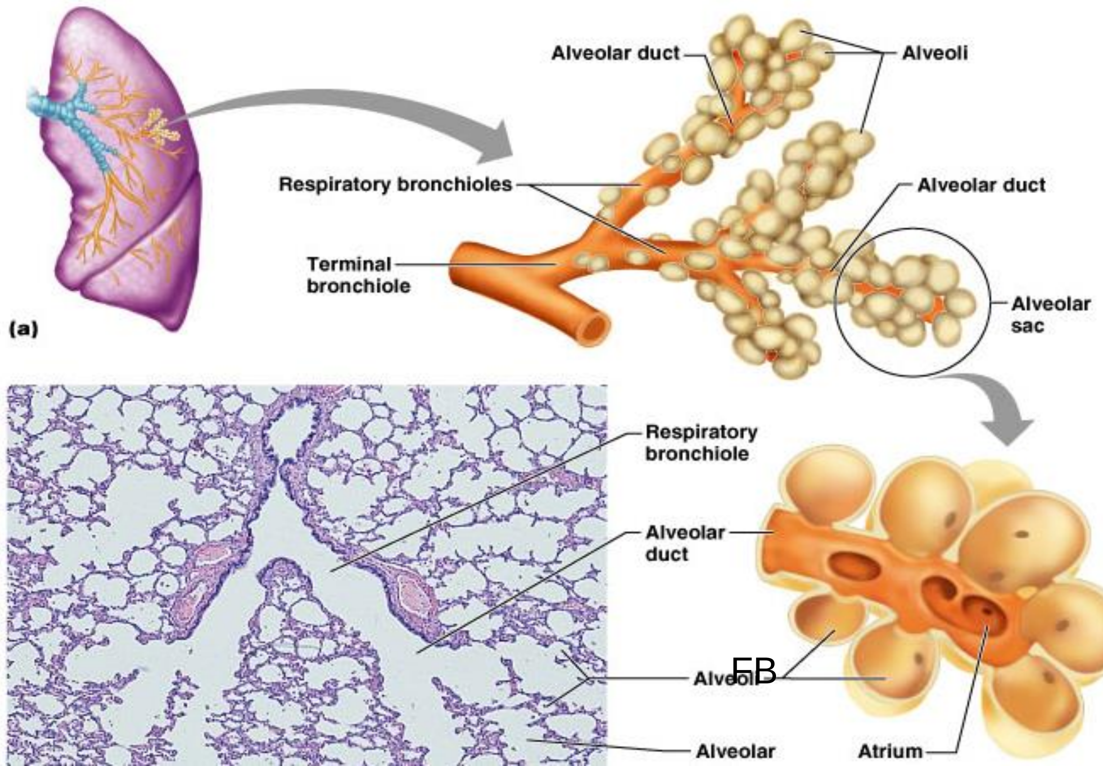
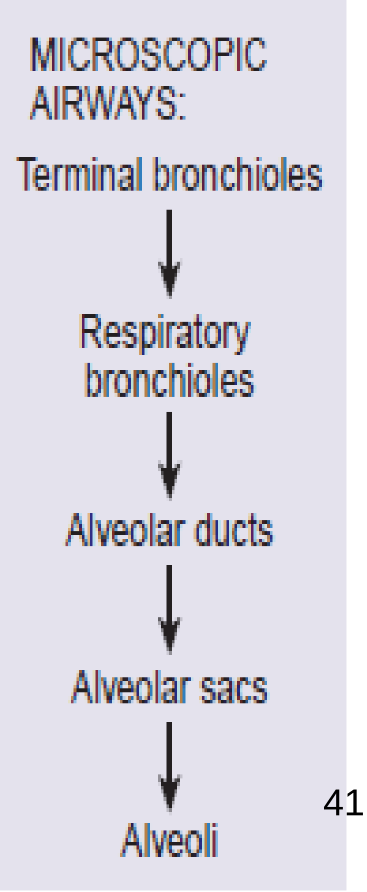
Respiratory zone

- Contain **alveoli :-**Structures that contain air-exchange chambers
- End-point of respiratory tree
- Respiratory bronchioles lead into alveolar ducts: walls consist of alveoli
- Ducts lead into terminal clusters called alveolar sacs – are microscopic chambers
- There are **3 million alveoli**!





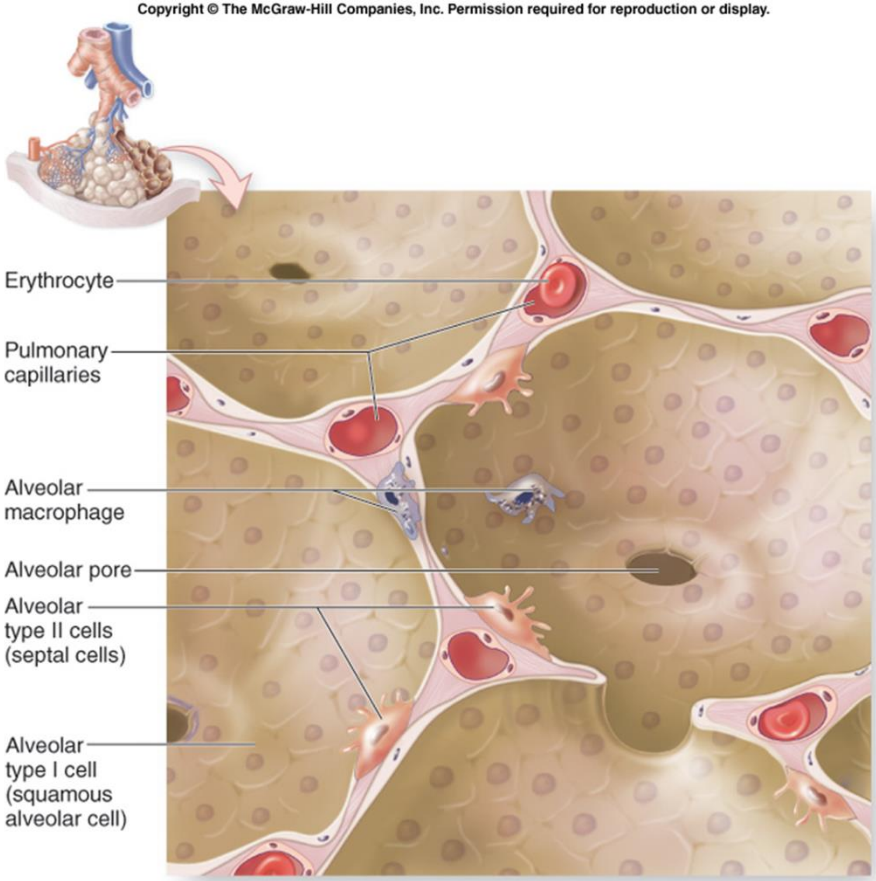
Cont….

❖ Cells of the interalveolar septum

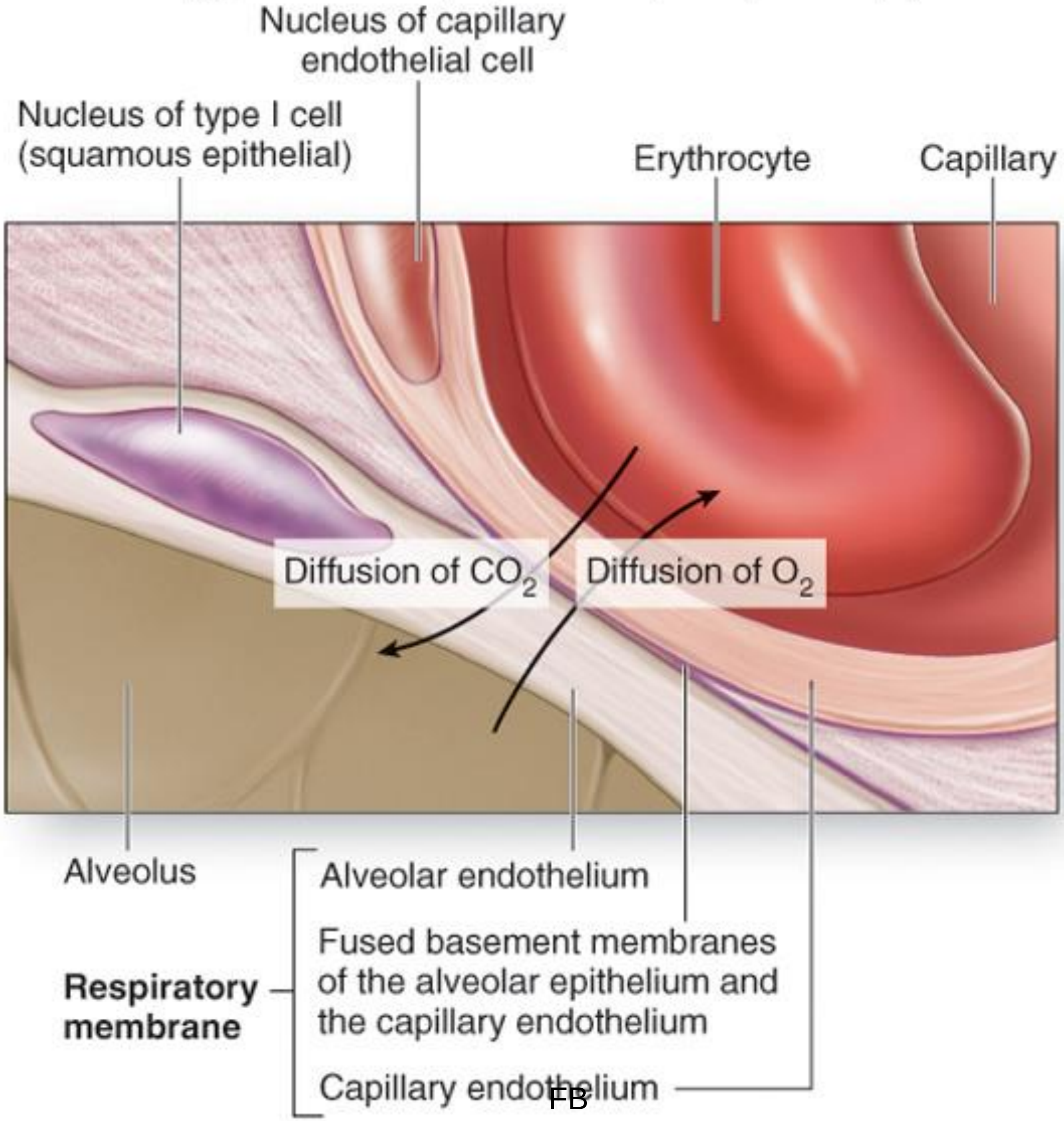
1. Capillary endothelial cells (30%)
2. Pneumocytes type I (8%)
3. Pneumocytes type II (Septal cells) (16%)
4. Interstitial cells (Fibroblast & mast cells) (36%)
5. Alveolar macrophages (Dust cells) (10%)

❖ Alveolar Pores

- They can **equalize the pressure inside the alveoli**
- They allow **collateral air circulation** if the bronchioles is obstructed



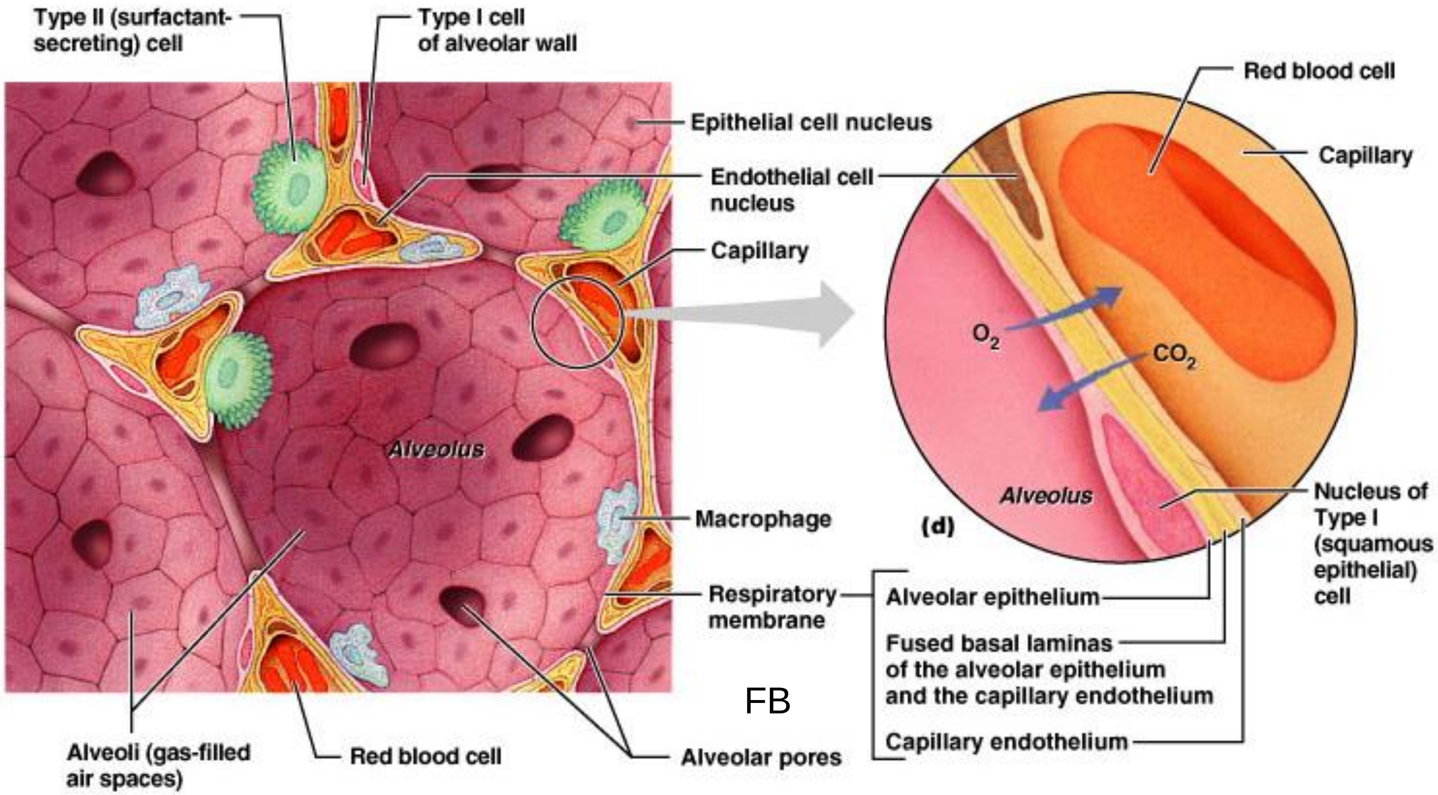
(a)


Nucleus of capillary endothelial cell
Nucleus of type I cell (squamous epithelial)
Erythrocyte
Capillary
Diffusion of CO_2
Diffusion of O_2
Alveolus
Alveolar endothelium
Fused basement membranes of the alveolar epithelium and the capillary endothelium
Respiratory membrane
Capillary endothelium

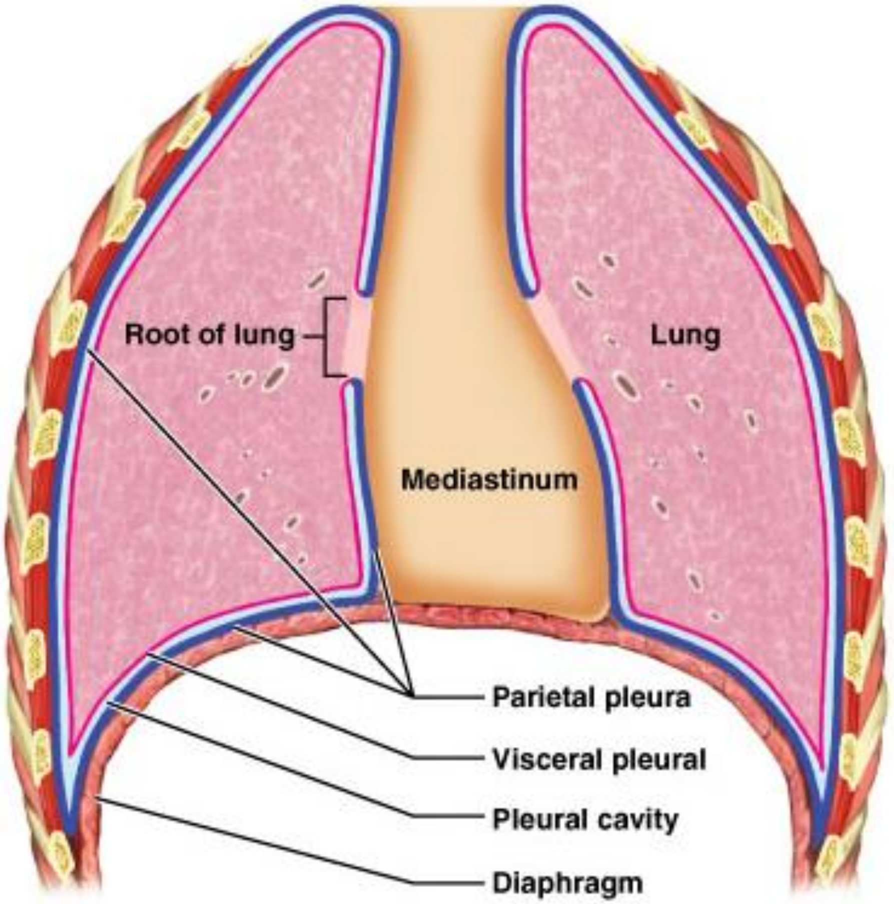
(b)

Surfactant

- **Type II** cuboidal epithelial cells are scattered in alveolar walls
- Surfactant is a detergent-like substance which is secreted in fluid coating alveolar surfaces – it decreases tension
- Without it the walls would stick together during exhalation



Pleura and lung



Around each lung is a flattened sac of serous membrane called ***pleura***

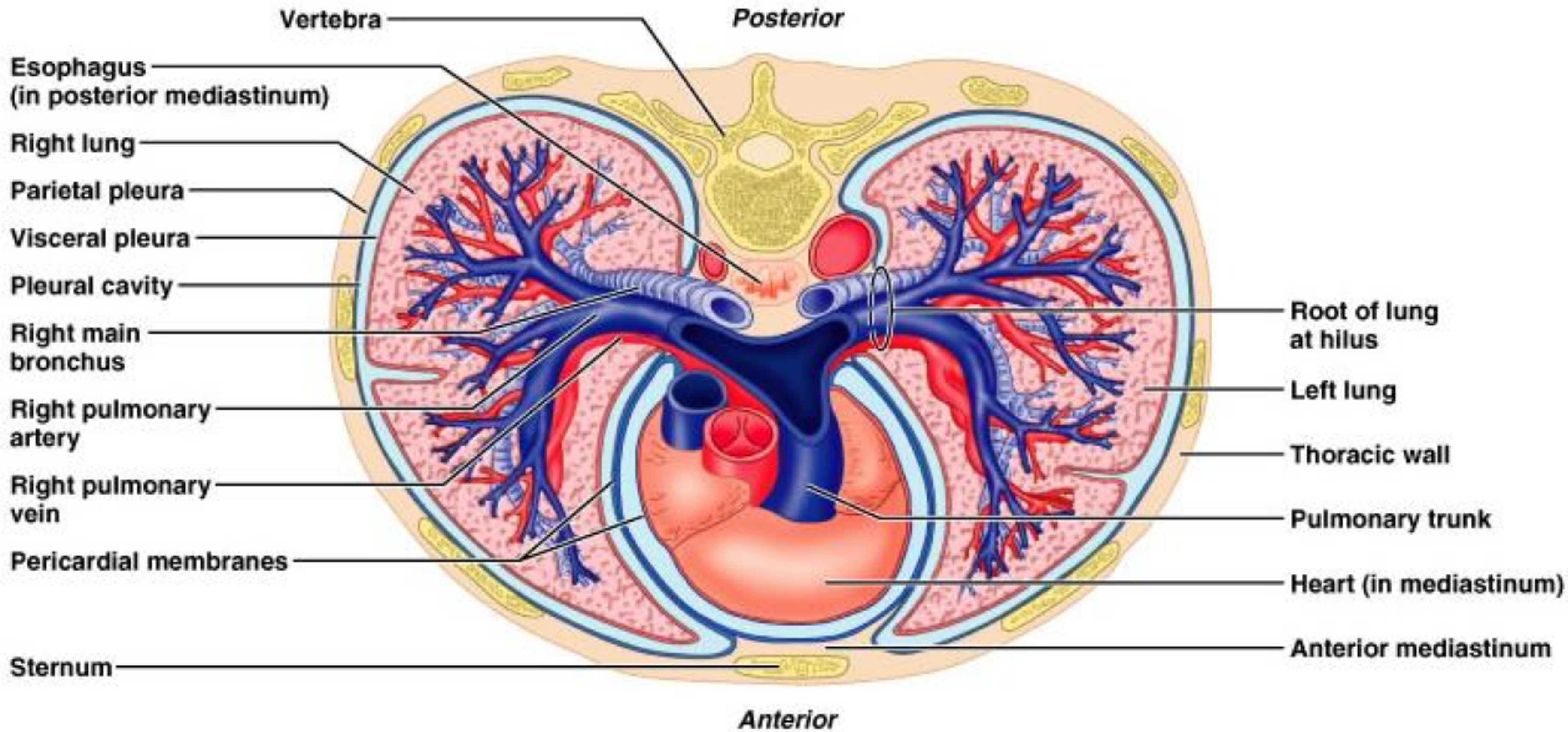
Layers

- ***a)Parietal pleura*** – outer layer
- ***b)Visceral pleura*** – directly on lung

Pleural cavity – slit-like potential space filled with pleural fluid

- Lungs can slide but separation from pleura is resisted (like film b/n 2 plates of glass)
- Lungs cling to thoracic wall & are forced to expand and recoil as volume of thoracic cavity changes during breathing

Relationship of organs in thoracic cavity



Lungs

- Each lung has a conical shape
- In the adult, the **right** lung is **700g** & the **left** lung is **600g** in weight

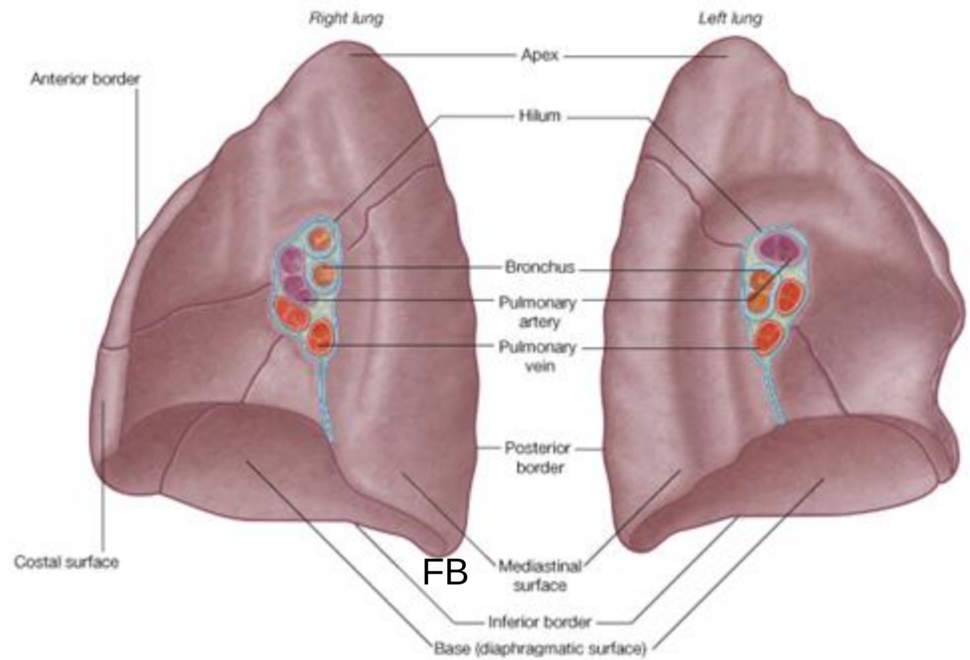
❑ **Features**

a) Surfaces:-costal, mediastinal & diaphragmatic

b) Borders:-anterior, posterior and inferior

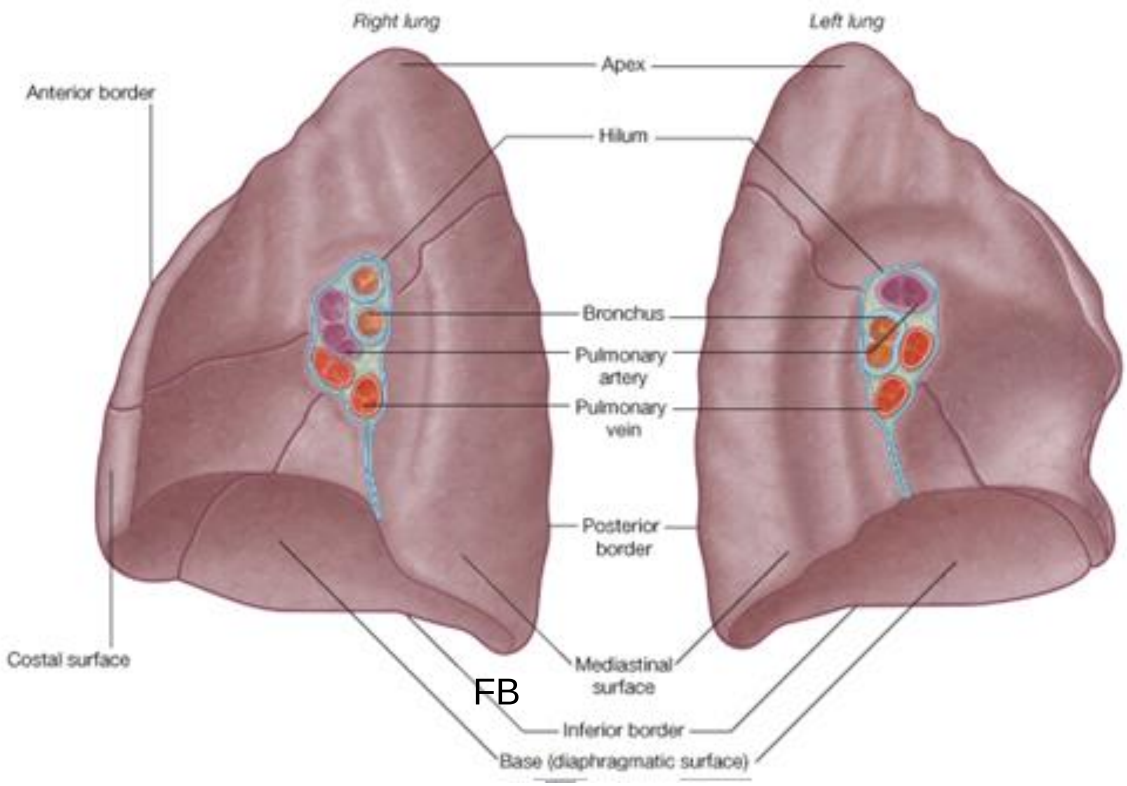
c) Apex:- is superior tip, just deep to clavicle

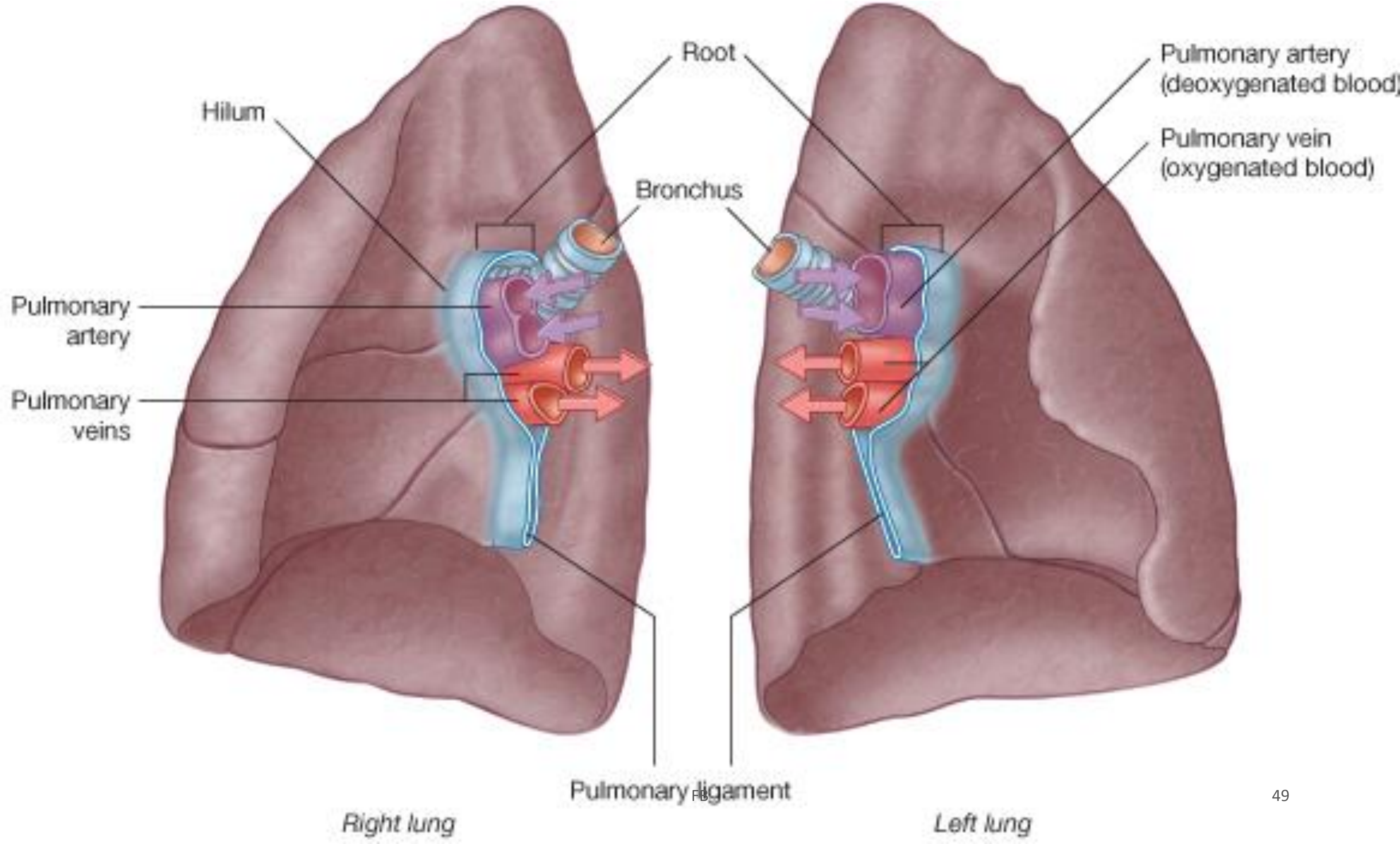
d) Base:- is concave inferior surface resting on diaphragm



Hilus or (hilum) of lung

- Indentation on mediastinal (medial) surface
- Place where blood vessels, bronchi, lymph vessel, & nerves enter and exit lung
- Main ones: **pulmonary artery & veins and main bronchus**
- The tubular sheath of pleura that surrounds the root of lung is called pulmonary ligament



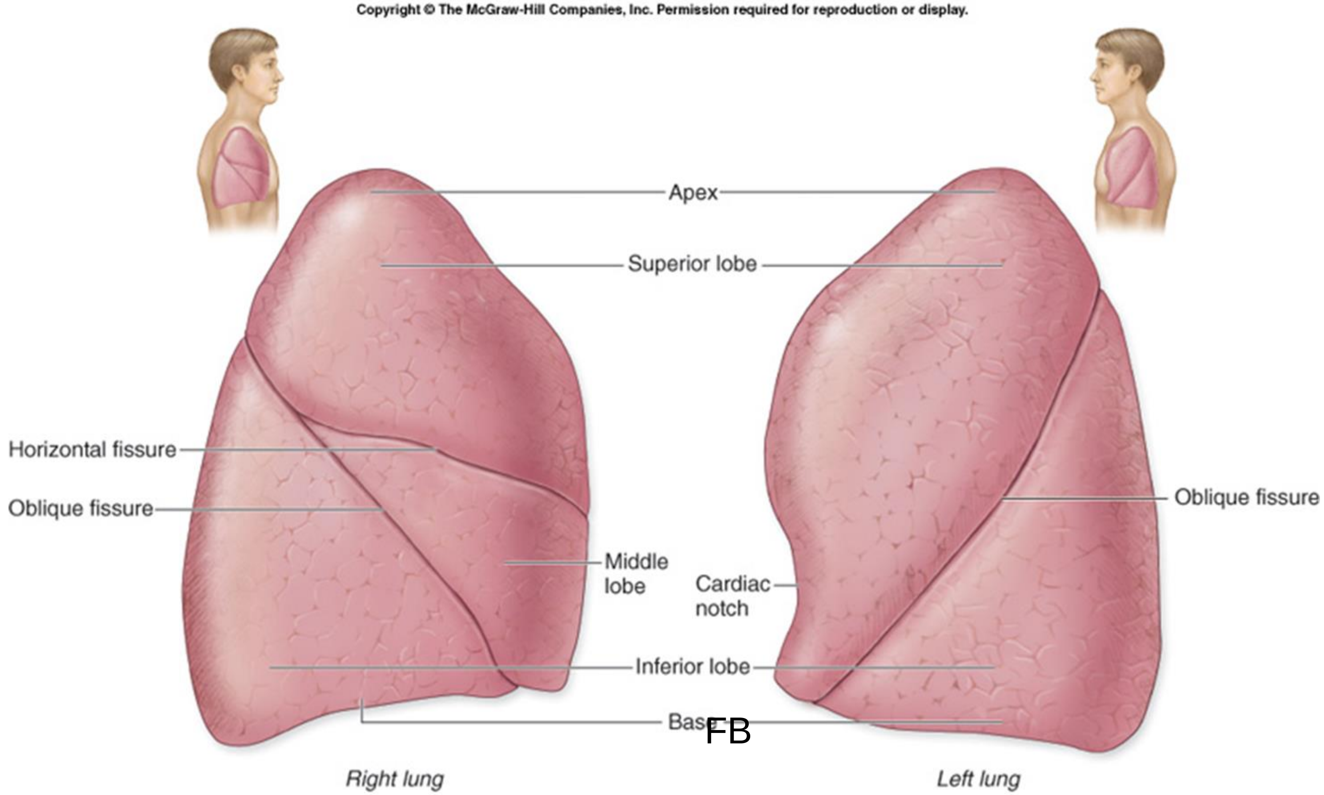

Root
Hilum
Pulmonary artery
(deoxygenated blood)
Pulmonary vein
(oxygenated blood)
Bronchus
Pulmonary
artery
Pulmonary
veins
Pulmonary ligament
Right lung
Left lung

Right lung

- 3 lobes
  a) Upper lobe
  b) Middle lobe
  c) Lower lobe
- 2 fissure (Horizontal & Oblique fissures)

Left lung

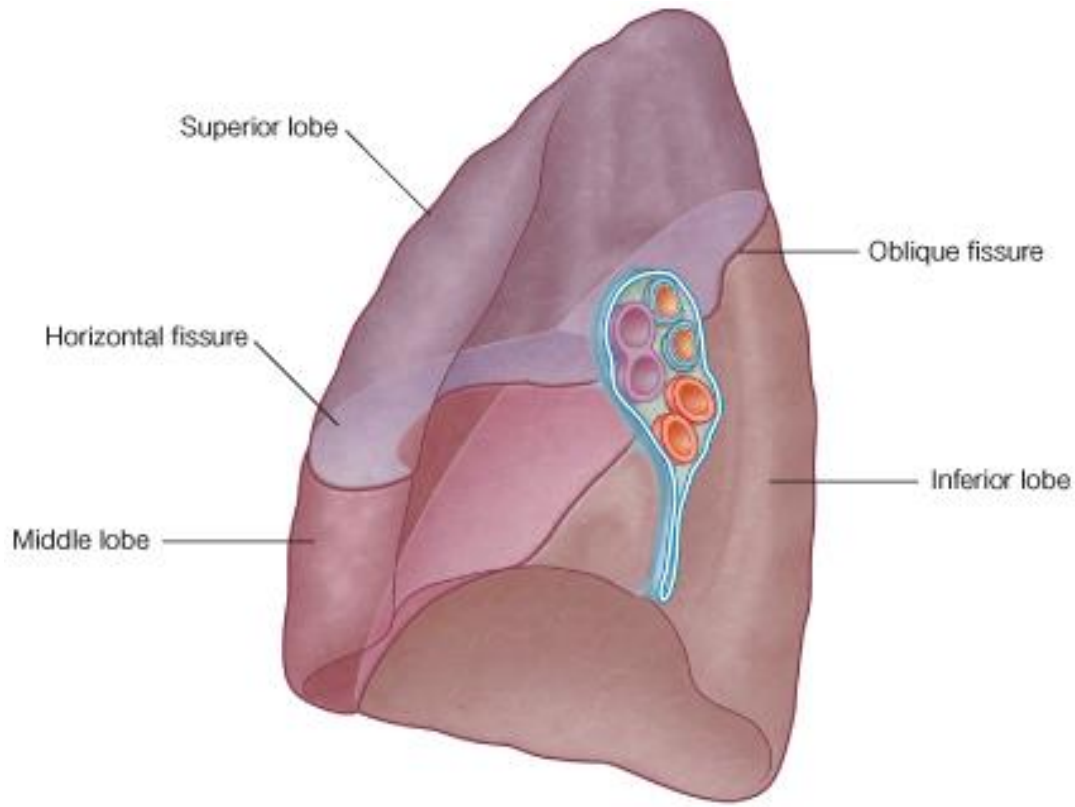
- 2 lobes
  a) Upper lobe
  b) Lower lobe
- 1 fissure (Oblique fissure)



(a) Lateral views

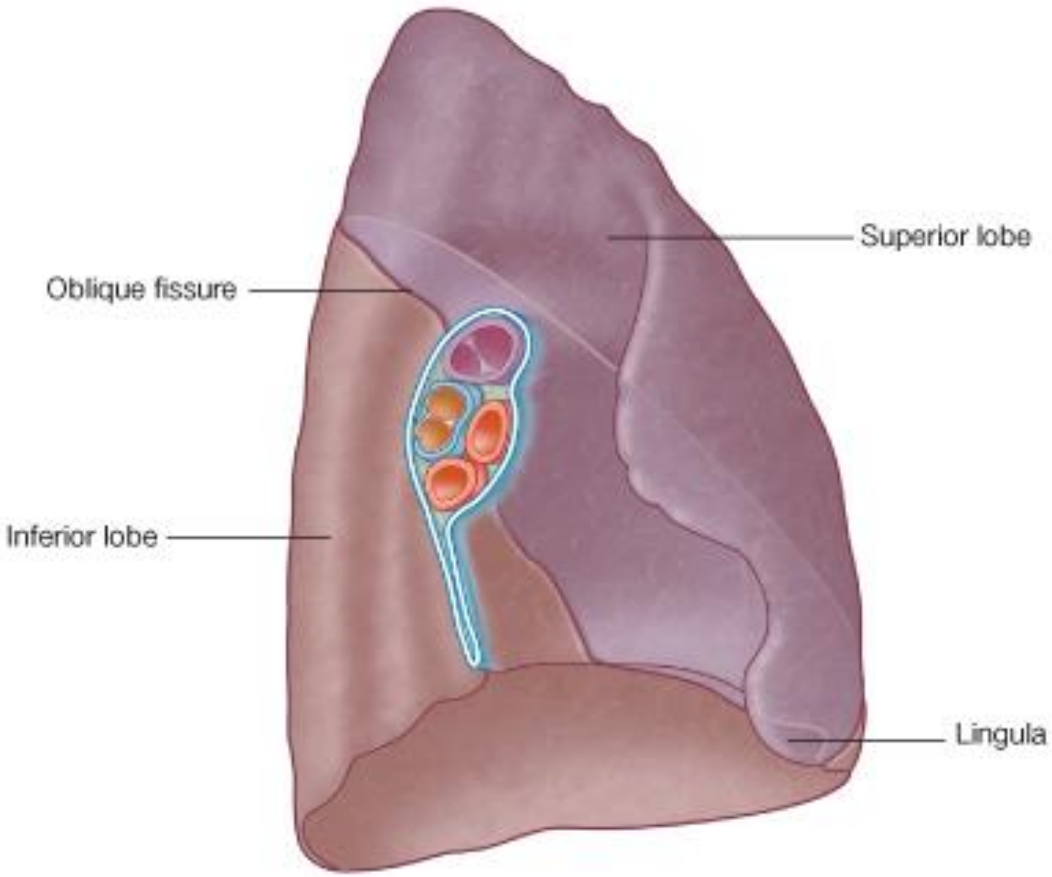
Right Lung

- The right lung is slightly **larger** than the left lung and is divided into three lobes by two fissures.
- **The oblique fissure**
  - ➢ It runs from the inferior border upward and backward across the medial and costal surfaces until it cuts the posterior border about 6 cm below the apex
- **The horizontal fissure**
  - ➢ It runs horizontally across the costal surface at the level of the fourth costal cartilage to meet the oblique fissure in the midaxillary line
  - ➢ The middle lobe is thus a small triangular lobe bounded by the horizontal and oblique fissures



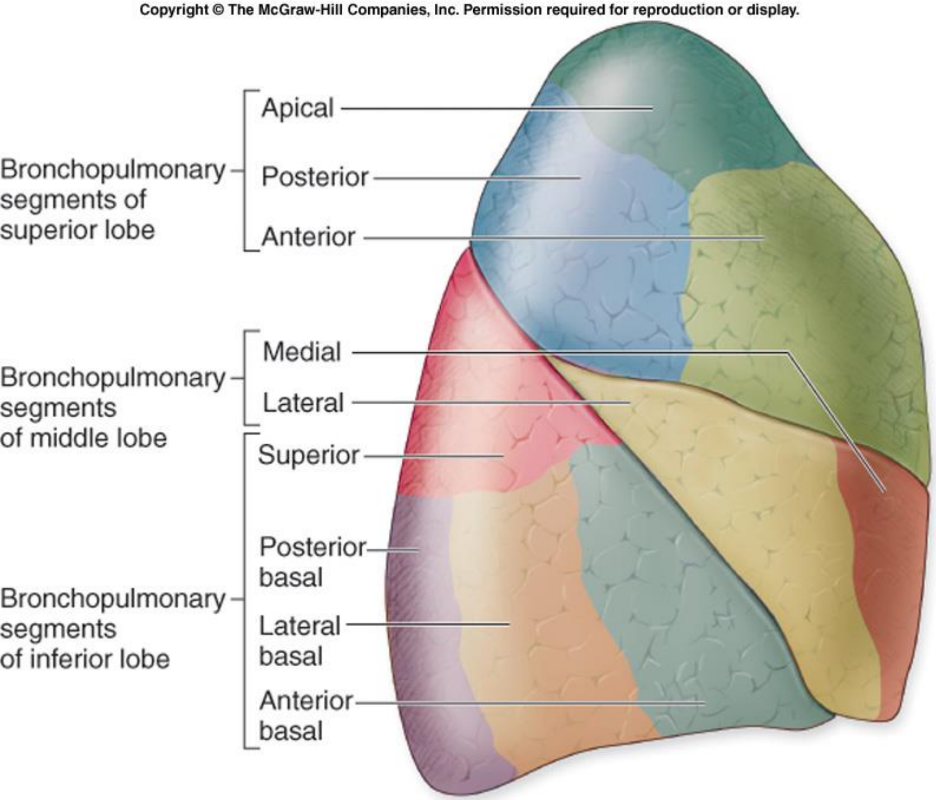
Left Lung

- The left lung is divided into two lobes by a **oblique fissure**
- The lobes are the upper and lower lobes
- There is **no horizontal fissure** in the left lung
- Just below the **cardiac notch** there is tongue like projection in the left lung & is called as the **lingula**



Cont….

- Each lobe is made up of ***bronchopulmonary segments*** separated by dense connective tissue
  - Each segment receives air from an individual ***segmental (tertiary) bronchus***
  - In each lung:-approximately 10 bronchopulmonary segments
  - Limit spread of infection
  - Can be removed more easily because only small vessels span segments
- Smallest subdivision seen with naked eye is ***lobule***
  - Hexagonal on surface, size of pencil eraser
  - Served by large bronchiole & its branches
  - Black carbon is visible on CT separating individual lobules in smokers & city dwellers



Right lung, lateral view

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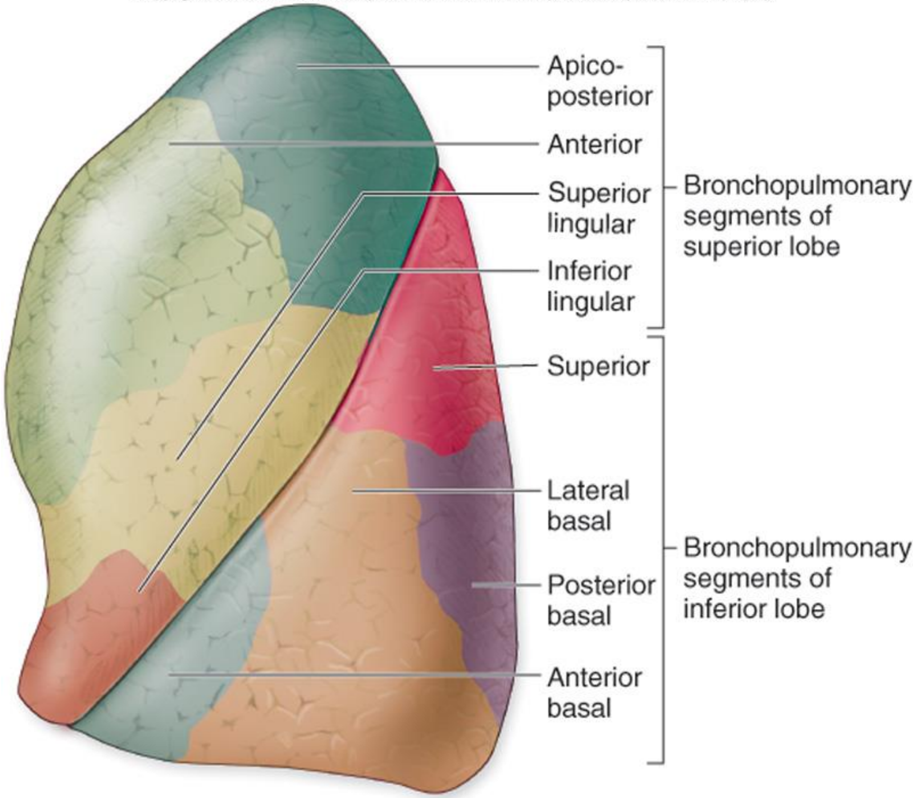
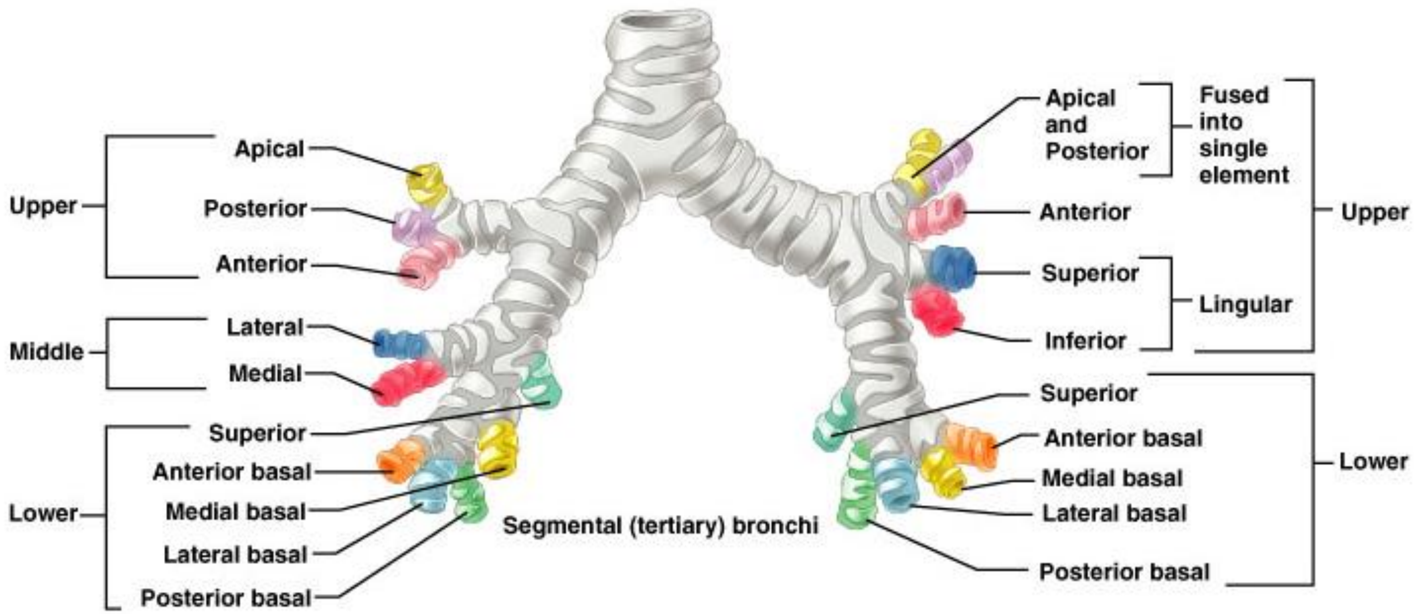


Left lung, lateral view



- Pulmonary arteries
  - bring oxygen-poor blood to lungs for oxygenation
  - They branch along with bronchial tree
  - The smallest feed into pulmonary capillary network around alveoli
- Pulmonary veins
  - carry oxygenated blood from alveoli of lungs to heart

- **Stroma**
  - framework of CT holding air tubes & spaces
  - Many elastic fibers
  - Lungs light, spongy & elastic
  - Elasticity reduces effort of breathing
- **Blood supply**
  - Lungs get their own blood supply from **bronchial arteries & veins**
- **Innervation**:
  - **pulmonary plexus** on lung root contains sympathetic, parasympathetic & visceral sensory fibers to each lung
    - From there, they lie on bronchial tubes & blood vessels within the lungs

Thank you